# UNIFORM SHEAFY TATE RINGS THAT ARE NOT STABLY UNIFORM

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Abstract. Buzzard–Verberkmoes and Mihara show that a stably uniform Tate Huber ring is sheafy. We construct examples of uniform and strongly sheafy Tate rings which are not stably uniform, answering Question 7 of Kedlaya's *Nonarchimedean Scottish Book*. In the first example a rational localisation is non-reduced, whilst in the second the rational localisation is an integral domain but is not uniform. The two main results are due to ChatGPT 5.6 Sol. Both constructions and the rational localisations witnessing failure of stable uniformity have been formalised in Lean 4; the strong-sheafiness formalisation is complete for the second example and for the Laurent-series case of the first.

## 1. Introduction

The examples of Buzzard–Verberkmoes and Mihara show that uniformity can be lost on a rational chart and, separately, that a uniform Tate Huber ring need not be sheafy [4, 18]. They leave open whether every uniform sheafy Tate Huber ring is stably uniform. Hansen and Kedlaya ask this explicitly [8, Remark 3.16]; it is also Question 7 of Kedlaya's *Nonarchimedean Scottish Book* [19]. We answer this question in the negative.

Following Hansen–Kedlaya [8, Definition 4.1], a Huber ring $R$ is *strongly sheafy* if $R\langle V_1, \dots, V_s\rangle$ is sheafy for every $s \geq 0$. In fact, both examples below are strongly sheafy.

The examples and proofs we will describe here were found with the help of ChatGPT 5.6 Sol. In Section 9 we describe how we used ChatGPT in the search for these examples. The Lean formalisation was carried out by Claude Code, and appendix A records the formal definitions and theorem statements. This article was prepared with the help of AI from an initial human-written skeleton.

Our first example is what the AI calls the "finite-jet pullback" example. This is a non-noetherian uniform integral domain for which a certain rational localisation is non-reduced, and hence is no longer uniform. Let $k$ be a complete discretely valued nonarchimedean field, with valuation ring $k^\circ$, uniformizer $\varpi$, and residue field $\widetilde{k}$. Put

$$L_0 = k^\circ\langle W, W^{-1}\rangle, \qquad B_0 = k^\circ\langle W, Q\rangle/(Q^2),$$

$$C_0 = L_0\langle Q\rangle, \qquad D_0 = L_0\langle Q\rangle/(Q^2),$$

where $k^\circ\langle W, W^{-1}\rangle$ means $k^\circ\langle W, V\rangle/(WV-1)$. There are natural maps $B_0 \to D_0$ and $C_0 \twoheadrightarrow D_0$. Define

$$A_0 = B_0 \times_{D_0} C_0, \qquad A = A_0[1/\varpi],$$

and endow $A$ with the topology defined by the ring of definition $A_0$.

The main result is the following.

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**Theorem 1.1.** *The ring $A$ is a complete uniform non-noetherian Tate $k$-algebra and an integral domain, with $A^\circ = A_0$. It is strongly sheafy; in particular, the Huber pair $(A, A^\circ)$ is sheafy. However,*

$$A\left\langle \frac{W}{\varpi} \right\rangle \cong k\langle X, Q\rangle/(Q^2), \qquad X = \frac{W}{\varpi},$$

*so $A$ is not stably uniform.*

The key element in this example is that after the rational localisation $A \to A\langle W/\varpi\rangle$, $Q^2$ vanishes whereas $Q$ does not.

This construction also gives a new example solving Problems 24 and 28 of the *Nonarchimedean Scottish Book*. We recall the two problems and give the argument in section 7. These results have also been formalised in Lean.

The proof uses Milnor squares of complete Tate rings. Recall that a cartesian square of rings with a surjective leg is called a Milnor square [17, Section 2]; the complete Tate setting is used by Kerz–Saito–Tamme in their work on analytic $K$-theory [14, Section 3].

**Definition 1.2.** A commutative square of complete Tate $k$-algebras

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D \end{array}$$

is a *strict Milnor square* if it is cartesian, the sequence of underlying topological $k$-vector spaces

$$0 \longrightarrow A \longrightarrow B \oplus C \xrightarrow{(b,c)\mapsto \bar{b}-\bar{c}} D \longrightarrow 0$$

is strict exact, and $C \to D$ is a strict surjection.

The maps $C \to D$ and $B \oplus C \to D$ are continuous surjections between complete normed $k$-vector spaces, and $A$ has the subspace topology as the kernel of the second. The nonarchimedean open mapping theorem [14, Lemma 3.1] therefore shows that the square defining $A$ is strict. Our main input is that, for this finite-jet square, the strict exact row persists after completed rational localisation and is compatible with refinement. This transfers sheafiness from $B, C, D$ to $A$, giving new ways for proving that these rings are sheafy. The second example shows that the failure of stable uniformity need not come from a non-reduced localisation.

**Remark 1.3** (Relation with earlier work)**.** It is difficult to understand exactly what sources an AI agent such as ChatGPT may have used in doing this work. In the interest of giving appropriate credit, we have listed here closely related work that may have been used in the development of the results here.

Buzzard–Verberkmoes construct a uniform affinoid ring with a rational localisation which is not uniform and, separately, a uniform affinoid ring whose structure presheaf is not a sheaf [4, Propositions 17 and 18]. Mihara gives further examples of both phenomena [18]. These examples do not give a uniform sheafy ring which is not stably uniform.

Ben-Bassat–Kremnizer prove strict two-term resolutions for Weierstrass and Laurent localisations, and reduce a general rational localisation to these cases [2, Lemmas 5.13–5.14]. Bambozzi–Kremnizer formulate rational localisation in terms of Koszul complexes and prove strict Koszul regularity [3, Notation 4.3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13]. The bounded-denominator estimates in Lemma 5.1 are the form needed to show that rational localisation preserves the finite-jet Milnor square. The resulting uniform sheafy domain which is not stably uniform appears to be new.

The weighted-parity construction of section 8 is closer to the infinite monomial-support examples of Buzzard–Verberkmoes and Mihara. Its additional feature is that rational localisation can be calculated in a noetherian finite-variable subring.

## 2. Conventions and notation

We follow Huber for Huber pairs, adic spectra, rational subsets, and rational localisation [11], and use Wedhorn's account for strong noetherianity and sheafiness [21, Proposition and Definition 6.36, Section 8.1]; the bounded-denominator language for strictness follows Buzzard–Verberkmoes [4], and the quotient presentation of a rational localisation is also discussed by Hansen–Kedlaya [8].

- *Tate topology.* Throughout, $E$ denotes a complete Tate $k$-algebra. In the discretely valued setting of this paper, this means that $E$ contains a topologically nilpotent unit and admits an open bounded $k^\circ$-subalgebra $E_0$ such that
$$E = E_0[1/\varpi], \qquad \{\varpi^n E_0\}_{n\geq 0}$$
is a basis of neighbourhoods of zero. Such an $E_0$ is called a *ring of definition*. We always choose it closed, hence $\varpi$-adically complete. A subset of $E$ is bounded if it is contained in $\varpi^{-r}E_0$ for some $r \geq 0$.
- *Uniformity and noetherianity.* We write $E^\circ$ for the ring of power-bounded elements. The ring $E$ is *uniform* if $E^\circ$ is bounded, and *strongly noetherian* if $E\langle Z_1, \ldots, Z_s\rangle$ is noetherian for every $s \geq 0$. A Tate ring is *stably uniform* if every rational localisation is uniform.
- *Huber pairs and sheafiness.* A *ring of integral elements* is an open subring $E^+ \subseteq E^\circ$ which is integrally closed in $E$. We call the Huber pair $(E, E^+)$ *sheafy* when its structure presheaf, valued in complete topological rings, satisfies the sheaf condition. Thus, for a finite rational cover $U = \bigcup_i U_i$, restriction identifies $\mathcal{O}(U)$ homeomorphically with the subspace of $\prod_i \mathcal{O}(U_i)$ consisting of tuples whose restrictions to every $U_i \cap U_j$ agree.
- *Strictness.* We use the following bounded-denominator form of strictness. Let $u : M \to N$ be a continuous $k$-linear map, and let $M_0 \subset M$ and $N_0 \subset N$ be open bounded $k^\circ$-submodules. Then $u$ is strict if and only if there exists $a \geq 0$ such that
$$\varpi^a\big(u(M) \cap N_0\big) \subset u(M_0).$$
For a surjection this becomes $\varpi^a N_0 \subset u(M_0)$. For an injection it says that the inverse image of $N_0$ is bounded in $M$. This is the criterion used by Buzzard–Verberkmoes [4, Lemma 2]. We use the term *strict Milnor square* in the sense of Definition 1.2; the underlying algebraic terminology is standard [17, 14].
- *Rational localisation.* We take every rational datum to have at least one numerator; an empty list may be padded by $f_1 = 0$. Let $\alpha = (f_1, \ldots, f_m; g)$ be a rational datum in $E$, so the ideal generated by $g, f_1, \ldots, f_m$ is open. Since $E$ is a Tate $k$-algebra, this ideal is the unit ideal. After multiplying the entire datum by a common power of $\varpi$, we may assume that all entries lie in $E_0$.

  The rational localisation $E_\alpha$ is the $\varpi$-adic completion of $E[1/g]$ for the ring of definition generated by
$$E_0\left[\frac{f_1}{g}, \ldots, \frac{f_m}{g}\right].$$
For such a ring of definition $R$, we use the notation
$$\widehat{R} = \varprojlim_{n\geq 1} R/\varpi^n R, \qquad \ker(R \longrightarrow \widehat{R}) = \bigcap_{n\geq 1} \varpi^n R.$$

$E_\alpha$ is independent of the ring of definition and of the presentation of the rational domain, and it is transitive under rational refinement. Put

$$T_E = E\langle T_1, \dots, T_m\rangle = E_0\langle T_1, \dots, T_m\rangle[1/\varpi], \qquad r_i = gT_i - f_i,$$

and

$$I_{E,\alpha} := (r_1, \dots, r_m) = \operatorname{im}\left(T_E^m \xrightarrow{d_{1,E}} T_E\right) \subset T_E, \qquad d_{1,E}(u_1, \dots, u_m) = \sum_i u_i(gT_i - f_i).$$

Then

$$E_\alpha \cong T_E/\overline{I_{E,\alpha}}. \tag{1}$$

Here the bar is necessary until the ideal has been shown to be closed. This is the standard quotient presentation of rational localisation; see [11, Notations, (1.2), Proposition 1.3, Lemma 1.5] and [8, Definition 3.6].

## 3. The finite-jet ring

In this section we identify the finite-jet algebra with a concrete subring of $C$ and prove that it is a complete uniform domain. Recall that

$$L_0 = k^\circ\langle W, W^{-1}\rangle, \qquad B_0 = k^\circ\langle W, Q\rangle/(Q^2),$$
$$C_0 = L_0\langle Q\rangle, \qquad D_0 = L_0\langle Q\rangle/(Q^2),$$

where $k^\circ\langle W, W^{-1}\rangle$ denotes $k^\circ\langle W, V\rangle/(WV-1)$. The natural maps $B_0 \to D_0$ and $C_0 \twoheadrightarrow D_0$ allow us to define

$$A_0 = B_0 \times_{D_0} C_0.$$

We write

$$A = A_0[1/\varpi], \quad L = L_0[1/\varpi], \quad B = B_0[1/\varpi], \quad C = C_0[1/\varpi], \quad D = D_0[1/\varpi].$$

Each of these Tate rings carries the topology defined by the corresponding subscript-zero ring.

The results in this section and the next do not require the valuation on $k$ to be discrete. They hold for any complete ultrametric normed field $k$ and any chosen $\varpi \in k^\times$ with $0 < |\varpi| < 1$, with the same definitions using the norm unit ball $k^\circ$. We return to the discretely valued hypothesis in section 5.

The quotient map $C_0 \to D_0$ has a continuous $k^\circ$-linear section

$$\sigma : D_0 \longrightarrow C_0, \qquad \sigma(f_0 + Qf_1) = f_0 + Qf_1,$$

obtained by viewing the unique representative of $Q$-degree at most one as an element of $C_0$. Consequently

$$0 \longrightarrow A_0 \longrightarrow B_0 \oplus C_0 \longrightarrow D_0 \longrightarrow 0$$

is exact, and $A_0$ is $\varpi$-adically complete. After inverting $\varpi$ we obtain a strict Milnor square

$$\begin{array}{ccc} A & \longrightarrow & C \\ \downarrow & & \downarrow \\ B & \longrightarrow & D. \end{array}$$

The map $B_0 \to D_0$ is injective. Projection to $C_0$ therefore identifies $A_0$ with the subring

$$A_0 = \left\{ f_0(W) + Qf_1(W) + Q^2 h(W, W^{-1}, Q) : f_0, f_1 \in k^\circ\langle W\rangle,\ h \in C_0 \right\}. \tag{2}$$

Equivalently, $A_0$ consists of the restricted series

$$\sum_{(a,b)\in S} c_{a,b} W^a Q^b, \qquad S = \{(a,b) \in \mathbb{Z} \times \mathbb{N} : b \le 1 \Rightarrow a \ge 0\},$$

with $c_{a,b} \in k^\circ$. Here restricted means that, for every $n$, only finitely many coefficients are nonzero modulo $\varpi^n$. The set $S$ is closed under addition, so this is a subring of $C_0$. The same description with coefficients in $k$ gives $A \subset C$.

Equip $A, B, C, D$ with the coefficientwise Gauss norms whose unit balls are $A_0, B_0, C_0, D_0$, respectively. Write

$$\iota_B : A \to B, \qquad \iota_C : A \to C, \qquad \rho_B : B \to D, \qquad \rho_C : C \to D$$

for the structural maps, and give $A$ the restricted Gauss norm from $C$. For $a = f_0 + Qf_1 + Q^2h$, the coefficient formulas give

$$\|\iota_C(a)\|_C = \|a\|_A, \qquad \|\iota_B(a)\|_B \le \|a\|_A, \qquad \|\rho_B(b)\|_D \le \|b\|_B, \qquad \|\rho_C(c)\|_D \le \|c\|_C. \quad (3)$$

Thus every structural map is norm-nonincreasing.

**Proposition 3.1.** *The ring $A$ is a complete uniform non-noetherian Tate $k$-algebra and an integral domain. Moreover,*

$$A^\circ = A_0.$$

*Proof.* The Gauss norm on $L = k\langle W, W^{-1}\rangle$ is multiplicative, and the restricted Gauss norm on $C = L\langle Q\rangle$ is therefore multiplicative as well. In particular, $C$ is a domain, and hence so is its subring $A$. The constant- and linear-coefficient maps $C \to L$ are continuous, and $k\langle W\rangle$ is closed in $L = k\langle W, W^{-1}\rangle$ for the Gauss norm. Hence the condition that these two coefficients lie in $k\langle W\rangle$ defines a closed subring of $C$, so $A$ is complete. The image of $\varpi$ in $A$ is a topologically nilpotent unit, and therefore $A$ is a Tate ring.

The coefficient description of $A_0$ shows that it is exactly the unit ball for the restricted Gauss norm on $A$. Since this norm is multiplicative,

$$\|a^n\| = \|a\|^n.$$

It follows that $a$ is power-bounded if and only if $\|a\| \le 1$. Thus $A^\circ = A_0$, and the unit ball is bounded, so $A$ is uniform.

To see that $A$ is not noetherian, let $J = \ker(\iota_B) = Q^2C \subset A$. If $J$ were generated by $a_1, \ldots, a_r$, write $b_i \in L$ for the coefficient of $Q^2$ in $a_i$. Now, for every $\ell \in L$, we can write

$$Q^2\ell = \sum_i x_i a_i, \qquad x_i \in A,$$

and write $c_i$ for the constant coefficient of $x_i$ as a series in $Q$. By (2), we have $c_i \in k\langle W\rangle$. Since $a_i \in Q^2C$, its coefficients in degrees 0 and 1 vanish. The coefficient of $Q^2$ in $x_i a_i$ is therefore exactly $c_i b_i$. Comparing the coefficients of $Q^2$ on both sides gives

$$\ell = \sum_i c_i b_i, \qquad c_i \in k\langle W\rangle.$$

Thus $b_1, \ldots, b_r$ would generate $L$ as a $k\langle W\rangle$-module. This is impossible. Indeed, it would make $W^{-1}$ integral over $k\langle W\rangle$. Multiplying a monic relation for $W^{-1}$ of degree $n$ by $W^n$ would put 1 in the proper ideal $(W) \subset k\langle W\rangle$. Thus $J$ is not finitely generated, and $A$ is not noetherian. □

## 4. A rational localisation which is not uniform

We next compute the rational chart which witnesses the failure of stable uniformity.

Consider the datum

$$\alpha = (W; \varpi).$$

The ideal $(\varpi, W)$ is open, since $\varpi$ is a unit in $A$. Let

$$A_\alpha = A\left\langle \frac{W}{\varpi} \right\rangle$$

be the corresponding rational localisation on which $|W| \leq |\varpi|$, and write $X = W/\varpi$. Since $\varpi$ is already invertible in $A$, the algebraic localisation does not change the ring. More explicitly, $A_\alpha$ is obtained by taking the $\varpi$-adic completion of $A_0[X] \subset A$ and then inverting $\varpi$.

**Proposition 4.1.** *There is a canonical isomorphism of complete topological rings*

$$A_\alpha \cong k\langle X, Q\rangle/(Q^2), \qquad W \longmapsto \varpi X, \quad Q \longmapsto Q.$$

*Proof.* Put

$$T_0 = k^\circ\langle X, Q\rangle/(Q^2), \qquad T = T_0[1/\varpi].$$

Let $\rho : A \to A_\alpha$ be the canonical map. We first show that

$$Q^2 C \subseteq \ker(\rho). \tag{4}$$

Take $y \in Q^2 C$, and choose $r \geq 0$ such that $\varpi^r y \in Q^2 C_0$. For every $n \geq 0$, the coefficient description of $A_0$ gives

$$W^{-n}\varpi^r y \in A_0.$$

Multiplication by $W^{-n}$ only introduces negative powers of $W$ in terms of $Q$-degree at least two. In $A_0[X]$, where $W = \varpi X$, we therefore have

$$\varpi^r y = W^n(W^{-n}\varpi^r y) = \varpi^n X^n (W^{-n}\varpi^r y) \in \varpi^n A_0[X]$$

for every $n$. Hence $\varpi^r y$ maps to zero in $\widehat{A_0[X]}$. Since $\varpi$ is invertible in $A_\alpha$, this proves (4).

We now construct the isomorphism. Recall that the map $\iota_B : A \to B$ from the defining pullback is given by

$$f_0(W) + Qf_1(W) + Q^2 h \longmapsto f_0(W) + Qf_1(W);$$

it simply discards the terms divisible by $Q^2$. The substitution $W \mapsto \varpi X$, $Q \mapsto Q$ defines a continuous homomorphism $B \to T$. Their composite is

$$\theta : A \longrightarrow T, \qquad f_0(W) + Qf_1(W) + Q^2 h \longmapsto f_0(\varpi X) + Qf_1(\varpi X).$$

It sends $A_0$ into $T_0$ and sends $W/\varpi$ to $X$. It therefore sends $A_0[X]$ into $T_0$ and extends, first to the completions and then after inverting $\varpi$, to a continuous homomorphism

$$\Phi : A_\alpha \longrightarrow T.$$

Set

$$\overline{X} = \rho(W)/\varpi, \qquad \overline{Q} = \rho(Q).$$

Both elements are power-bounded: they belong to the completed ring of definition coming from $A_0[X]$. Moreover, $\overline{Q}^2 = 0$ by (4). Evaluation at $\overline{X}$ therefore gives a continuous homomorphism

$$\Psi : T \longrightarrow A_\alpha, \qquad f + Qg \longmapsto f(\overline{X}) + g(\overline{X})\overline{Q}.$$

It remains to check the two composites. Write $a = f_0(W) + Qf_1(W) + y \in A$, with $y \in Q^2 C$. Then

$$(\Psi \circ \Phi)(\rho(a)) = f_0(\varpi\overline{X}) + \overline{Q} f_1(\varpi\overline{X}) = \rho\big(f_0(W) + Qf_1(W)\big) = \rho(a),$$

where the last equality uses (4). The image of $A$ is dense in $A_\alpha$, so continuity gives $\Psi \circ \Phi = 1$. Conversely, $\Phi \circ \Psi$ fixes $k$, $X$ and $Q$, and hence is the identity on the dense subring $k[X, Q]/(Q^2)$ of $T$. It follows by continuity that $\Phi \circ \Psi = 1$. Thus $\Phi$ and $\Psi$ are inverse continuous homomorphisms. $\square$

**Proposition 4.2.** *The ring $A$ is not stably uniform.*

*Proof.* In $T = k\langle X, Q\rangle/(Q^2)$, every element of the line $kQ$ is nilpotent and hence power-bounded. This line is not bounded with respect to the ring of definition $T_0$: for every $r \geq 0$,

$$\varpi^r\left(\varpi^{-(r+1)}Q\right) = \varpi^{-1}Q \notin T_0.$$

Thus $T$ is not uniform. By Proposition 4.1, $A_\alpha$ is not uniform. □

## 5. Localisation of the Milnor square

In this section we prove that rational localisation preserves the defining pullback. For the application to the finite-jet ring we again assume that $k$ is complete and discretely valued. We first record the strict Koszul estimate needed for the proof in the greater generality in which it was formalised.

5.1. **Koszul complexes for rational localisations.** Let $E$ be a complete ultrametric normed commutative ring with $\|1\| = 1$, and let $t \in E^\times$ satisfy

$$0 < \|t\| < 1, \qquad \|tx\| = \|t\|\|x\| \quad (x \in E).$$

Write $E_0 = \{x \in E : \|x\| \leq 1\}$ for its norm unit ball. Assume that $E_0$, $E\langle T_1, \ldots, T_m\rangle$, and the norm unit ball of this Tate algebra are noetherian. Let $f_1, \ldots, f_m, g \in E$ generate the unit ideal in $E$. Put

$$T_E = E\langle T_1, \ldots, T_m\rangle, \qquad T_{E,0} = E_0\langle T_1, \ldots, T_m\rangle,$$

and set $r_i = gT_i - f_i$.

We give $T_E$ the topology for which $\{t^n T_{E,0}\}_{n\geq 0}$ is a basis of neighbourhoods of zero. In the standard exterior basis, $\bigwedge^q T_E^m$ has the finite product topology, and each image has the subspace topology. Thus a differential is strict precisely when it is open onto its image.

The strict Koszul regularity theorem of Bambozzi–Kremnizer [3, Notation 4.3, Definitions 4.5–4.6, Proposition 4.9, Lemma 4.11, and Corollary 4.13] gives the exactness, strictness, and closed-image assertions below. We include the direct proof followed in the formalisation because the two bounds (5) and (6) are not stated there and are needed below.

**Lemma 5.1.** *The Koszul complex*

$$0 \longrightarrow \bigwedge^m T_E^m \longrightarrow \cdots \longrightarrow \bigwedge^2 T_E^m \xrightarrow{d_{2,E}} T_E^m \xrightarrow{d_{1,E}} T_E$$

*is exact in positive degrees. Every differential is continuous and strict onto its image, and every image is closed. In particular, if*

$$I_E = (r_1, \ldots, r_m) = \operatorname{im}(d_{1,E}) \subset T_E, \qquad d_{1,E}(u_1, \ldots, u_m) = \sum_i u_i r_i,$$

*then $I_E$ is closed. Moreover, there exists $h_E \in \mathbb{Z}_{\geq 0}$ such that*

$$t^{h_E}(I_E \cap T_{E,0}) \subset d_{1,E}(T_{E,0}^m). \tag{5}$$

*There also exists $z_E \in \mathbb{Z}_{\geq 0}$ such that*

$$t^{z_E}\left(\ker(d_{1,E}) \cap T_{E,0}^m\right) \subset d_{2,E}\left(\textstyle\bigwedge^2 T_{E,0}^m\right). \tag{6}$$

*Proof.* Over the polynomial ring $E[T_1, \ldots, T_m]$, the sequence $gT_i - f_i$ has Koszul complex exact in positive degrees. Indeed, this may be checked after localising at a prime. If one of the $gT_i - f_i$ is not in the prime, it is a unit after localisation and the Koszul complex is contractible. If all of them lie in the prime, then $g$ does not: otherwise all the $f_i$ would also lie in the prime, contrary to $(g, f_1, \dots, f_m) = E$. Thus $g$ is invertible there and

$$gT_i - f_i = g(T_i - f_i/g).$$

The elements $T_i - f_i/g$ form a regular sequence, since they are monic linear polynomials in distinct variables. This proves the assertion over the polynomial ring.

The map

$$E[T_1, \dots, T_m] \longrightarrow T_E$$

is flat. Indeed, $E_0$ is noetherian by assumption, so $E_0[T_1, \dots, T_m]$ is noetherian and its $t$-adic completion is flat over it [20, Tag 00MB]. After inverting $t$, the resulting map $E[T_1, \dots, T_m] \to T_E$ is flat. Tensoring the polynomial Koszul complex with $T_E$, and using the standard base-change identification of its finite free terms, proves positive-degree exactness. Notice that this does not assert exactness in degree zero.

Each differential is continuous because its coordinates are finite sums of multiplication maps. The image of $d_{1,E}$ is closed by the noetherian closed-ideal theorem. In higher degree, exactness identifies an image with the kernel of the next continuous differential, and hence that image is closed. The nonarchimedean open mapping theorem therefore shows that, for each differential $d_{q,E}$, there is a constant $C_{q,E} \geq 1$ such that every $y \in \mathrm{im}(d_{q,E})$ has a preimage $x$ satisfying

$$d_{q,E}(x) = y, \qquad \|x\| \leq C_{q,E}\|y\|.$$

This also proves that each differential is strict onto its image.

Choose $h_E \geq 0$ such that $\|t\|^{h_E} C_{1,E} \leq 1$. If $x \in I_E \cap T_{E,0}$, choose $u \in T_E^m$ with

$$d_{1,E}(u) = x, \qquad \|u\| \leq C_{1,E}\|x\| \leq C_{1,E}.$$

Then $t^{h_E} u \in T_{E,0}^m$ and $d_{1,E}(t^{h_E} u) = t^{h_E} x$, which proves (5). Similarly, choose $z_E \geq 0$ such that $\|t\|^{z_E} C_{2,E} \leq 1$. If $u \in \ker(d_{1,E}) \cap T_{E,0}^m$, choose $v$ with $d_{2,E}(v) = u$ and $\|v\| \leq C_{2,E}\|u\|$. Then $t^{z_E} v \in \bigwedge^2 T_{E,0}^m$, proving (6). ☐

5.2. **The defining ideals under pullback.** The formal result used here is not specific to the finite-jet rings. Let

$$\begin{array}{ccc} A & \xrightarrow{\phi_C} & C \\ {\scriptstyle \phi_B}\downarrow & & \downarrow{\scriptstyle \psi_C} \\ B & \xrightarrow{\psi_B} & D \end{array}$$

be a cartesian square of complete ultrametric normed commutative rings. We assume that $\phi_C$ and $\psi_B$ preserve the norm, that $\phi_B$ and $\psi_C$ do not increase it, and that

$$\|a\| = \max\{\|\phi_B(a)\|, \|\phi_C(a)\|\}.$$

We also assume that $\psi_C$ has a norm-preserving section as a map of sets, that $B, C, D$ have norm-scaling topologically nilpotent units as in Lemma 5.1. For the present number of variables, assume that the three Tate algebras $T_B, T_C, T_D$ and their norm unit balls are noetherian, and that the norm unit ball of $D$ is noetherian. These are precisely the noetherian inputs used in the formalisation; no noetherianity of the norm unit balls of $B$ and $C$ is required here. The finite-jet square satisfies them, with the coefficientwise section described in section 3 and with $t = \varpi$ at each vertex.

Fix elements

$$\alpha = (f_1, \dots, f_m; g)$$

in $A$ which generate the unit ideal. In the Tate setting this is a rational datum. Put

$$r_i = gT_i - f_i \in T_A,$$

and use the same notation for its image in $T_B, T_C$, and $T_D$. For $E \in \{A, B, C, D\}$ put

$$I_E = (r_1, \dots, r_m) = \mathrm{im}(d_{1,E}) \subset T_E.$$

Write $E_\alpha = T_E/I_E$ for the graph quotient. When the vertices are Tate rings, this is the completed rational localisation by (1).

The cartesian property and the section of $C \to D$, applied coefficientwise, give an exact sequence

$$0 \longrightarrow T_{A,0} \longrightarrow T_{B,0} \oplus T_{C,0} \longrightarrow T_{D,0} \longrightarrow 0. \tag{7}$$

The map $T_{C,0} \to T_{D,0}$ is surjective, as are the induced maps on finite free modules and their exterior powers. Exactness is coefficientwise: a compatible restricted coefficient pair lifts to $A_0$, and the max-norm identity ensures that the lifted coefficients still tend to zero.

**Lemma 5.2.** *The ideal $I_A$ is closed in $T_A$, and the canonical map*

$$I_A \longrightarrow I_B \times_{I_D} I_C$$

*is an isomorphism. Give each $I_E$ the subspace topology inherited from $T_E$, and give $I_B \oplus I_C$ the max norm. Then the sequence of topological abelian groups*

$$0 \longrightarrow I_A \longrightarrow I_B \oplus I_C \longrightarrow I_D \longrightarrow 0 \tag{8}$$

*is strict exact. Here the maps are*

$$x \longmapsto (x_B, x_C), \qquad (x_B, x_C) \longmapsto (x_B)_D - (x_C)_D.$$

*Proof.* For $E = B, C, D$, let $C_{1,E}$ be as in the proof of Lemma 5.1, and let $C_{2,D}$ be the corresponding constant for $d_{2,D}$. Take $(x_B, x_C) \in I_B \times_{I_D} I_C$, and choose $u_B \in T_B^m$ and $u_C \in T_C^m$ such that

$$d_{1,B}(u_B) = x_B, \qquad \|u_B\| \le C_{1,B}\|x_B\|,$$
$$d_{1,C}(u_C) = x_C, \qquad \|u_C\| \le C_{1,C}\|x_C\|.$$

In $T_D^m$ the difference

$$w = (u_B)_D - (u_C)_D$$

belongs to $\ker(d_{1,D})$. Choose $v_D \in \bigwedge^2 T_D^m$ such that

$$d_{2,D}(v_D) = w, \qquad \|v_D\| \le C_{2,D}\|w\|.$$

The norm-preserving coefficientwise section $T_D \to T_C$ induces a norm-preserving section on second exterior powers; use it to obtain $v_C \in \bigwedge^2 T_C^m$ mapping to $v_D$. Replacing $u_C$ by

$$u'_C = u_C + d_{2,C}(v_C)$$

does not change $d_{1,C}(u_C)$, and the images of $u_B$ and $u'_C$ in $T_D^m$ now agree. Exactness of (7), applied coefficientwise, gives $u_A \in T_A^m$ mapping to this pair. Then $d_{1,A}(u_A)$ maps to $(x_B, x_C)$.

These bounds, continuity of $d_{2,C}$, and the norm-preserving coefficientwise section give a constant $C$, independent of $(x_B, x_C)$, for which $x_A = d_{1,A}(u_A)$ satisfies

$$\|x_A\| \le C \max\{\|x_B\|, \|x_C\|\}.$$

The map $T_A \to T_B \oplus T_C$ is injective, so this proves both the algebraic pullback assertion and the boundedness of its inverse.

It remains to see that $I_A$ is closed. The ideals $I_B$ and $I_C$ are closed by Lemma 5.1. If $x$ lies in the closure of $I_A$, its images lie in $I_B$ and $I_C$. The pullback isomorphism just proved gives $x_A \in I_A$ with the same two images as $x$. Injectivity of $T_A \to T_B \oplus T_C$ gives $x = x_A$, as required.

Finally, for $y \in I_D$ choose $u_D \in T_D^m$ such that

$$d_{1,D}(u_D) = y, \qquad \|u_D\| \le C_{1,D}\|y\|.$$

Apply the norm-preserving section $T_D^m \to T_C^m$ and then $d_{1,C}$. This gives $x_C \in I_C$ mapping to $y$, with $\|x_C\|$ bounded by a fixed multiple of $\|y\|$. Thus $I_B \oplus I_C \to I_D$ is a strict surjection. Its kernel is $I_A$ by the pullback assertion, and the preceding bound for the inverse $I_B \times_{I_D} I_C \to I_A$ makes the inclusion on the left strict. This proves (8). □

**Proposition 5.3.** *For every unit-ideal datum $\alpha$ as above, the canonical sequence of graph quotients*

$$0 \longrightarrow A_\alpha \longrightarrow B_\alpha \oplus C_\alpha \longrightarrow D_\alpha \longrightarrow 0$$

*is strict exact, and $C_\alpha \to D_\alpha$ is a strict surjection. Equivalently,*

$$A_\alpha \cong B_\alpha \times_{D_\alpha} C_\alpha.$$

*Proof.* The ideals $I_B, I_C, I_D$ are closed by Lemma 5.1, and $I_A$ is closed by Lemma 5.2. Thus

$$E_\alpha = T_E/I_E \qquad (E = A, B, C, D).$$

We first prove algebraic exactness. If a representative $p \in T_A$ maps into both $I_B$ and $I_C$, the pullback assertion in Lemma 5.2 gives an element of $I_A$ with the same two images. Injectivity of $T_A \to T_B \oplus T_C$ then shows that $p \in I_A$. This gives injectivity on the left. Conversely, take compatible classes in $T_B/I_B$ and $T_C/I_C$ and choose representatives $b$ and $c$. Their defect in $T_D$ belongs to $I_D$. Lift this defect to an element of $I_C$ using the strict surjection in (8), and add it to $c$. The corrected representatives agree in $T_D$, so (7) lifts them to $T_A$. This proves algebraic exactness.

For the topology on the left, let $x \in A_\alpha$ and put

$$M = \max\{\|x_B\|, \|x_C\|\}.$$

Choose representatives $b \in T_B$ and $c \in T_C$ whose norms are arbitrarily close to the quotient norms of $x_B$ and $x_C$. Their defect lies in $I_D$. The norm estimate for $I_C \to I_D$ proved in Lemma 5.2 gives a constant $C_0$ and a correction $y \in I_C$ mapping to this defect such that

$$\|y\| \leq C_0 \|b_D - c_D\|.$$

Thus $b$ and $c + y$ agree in $T_D$ and lift to $p \in T_A$. The max-norm identity for the ambient pullback gives, after letting the error in the representatives tend to zero,

$$\|x\| \leq CM$$

for a constant $C$ depending only on the datum. The two quotient maps are norm-nonincreasing, so this estimate says precisely that $A_\alpha \to B_\alpha \oplus C_\alpha$ is a topological embedding.

The coefficientwise section $T_D \to T_C$ shows directly that $C_\alpha \to D_\alpha$ is surjective. More precisely, choose a representative in $T_D$ whose norm is arbitrarily close to the quotient norm of a given class in $D_\alpha$, and apply the norm-preserving section. Its class in $C_\alpha$ has no larger norm. Thus the surjection is open; together with continuity, this says that it is strict.

□

Applying the preceding two results to the finite-jet square gives the strict localised Milnor rows used below. For this square, the coefficient maps in the identifications commute with rational restriction; this is proved separately from presentation independence and transitivity. The same abstract result applies after adjoining any finite number of Tate variables.

## 6. Milnor descent and strong sheafiness

We first state the general descent result proved in the formalisation. It is a result about local section rings attached to rational-localisation data, rather than about an arbitrary cartesian square of global rings.

**Theorem 6.1** (Milnor descent)**.** *Let $R, B, C, D$ be the underlying rings of complete nonarchimedean Huber pairs, with continuous maps from $R$ to $B$, $C$, and $D$, and from $B$ and $C$ to $D$. Suppose that rational-localisation data on $R$ can be carried functorially to the other three vertices, that the resulting maps on section rings commute, and that this construction transports rational covers, is natural under refinement, and carries matching families to compatible families on common refinements. Suppose also that for every rational-localisation datum $U$ the map*

$$\mathcal{O}_R(U) \longrightarrow \mathcal{O}_B(U_B) \oplus \mathcal{O}_C(U_C)$$

*is a topological embedding whose image is the equaliser of the two maps to $\mathcal{O}_D(U_D)$, compatibly with restriction and matching families. If $B, C, D$ are sheafy, then $R$ is sheafy.*

*Proof.* Push a finite rational cover of $U$ to the other vertices. Separation over $B$ and $C$, followed by injectivity of the map in the theorem, gives separation over $R$. A matching family glues over $B$ and $C$; the two sections have the same image over $D$, and the equaliser hypothesis gives the unique section over $R$.

For the topology, the restriction map for $R$ factors, through the topological embedding in the theorem, into the product of the restriction maps for $B$ and $C$. Those maps are topological embeddings because $B$ and $C$ are sheafy. The factorisation therefore shows that the restriction map for $R$ is also a topological embedding. This is the topological sheaf condition on finite rational covers; the rational-basis theorem gives the sheaf condition on all opens [20, Tag 009O]. □

**Corollary 6.2.** *For every $s \geq 0$ and every ring of integral elements*

$$A_s^+ \subseteq A\langle V_1, \dots, V_s\rangle^\circ,$$

*the Huber pair*

$$\big(A\langle V_1, \dots, V_s\rangle, A_s^+\big)$$

*is sheafy. In particular, $A$ is strongly sheafy.*

The proof below works over the complete discretely valued base fixed in the introduction. The pinned Lean formalisation of this strong form currently covers the Laurent-series model over $F((t))$; ordinary sheafiness is formalised over an arbitrary complete discretely valued base.

*Proof.* The estimates (3) make the structural maps morphisms of Huber pairs. The rings $B, C, D$ are strongly noetherian affinoid $k$-algebras and hence sheafy by Huber's theorem [11, Theorem 2.2]. Proposition 5.3, together with its compatibility under refinement, supplies the hypotheses of Theorem 6.1. This proves sheafiness for $A$, independently of the choice of its ring of integral elements.

After adjoining $V_1, \dots, V_s$, the coefficientwise split square and the localised equaliser rows persist, while the other three vertices remain strongly noetherian. Applying the same theorem proves the assertion for every $s$ and every $A_s^+$. □

*Proof of Theorem 1.1.* Proposition 3.1 proves that $A$ is complete, uniform, non-noetherian, and a domain. Strong sheafiness is Corollary 6.2, and failure of stable uniformity is Proposition 4.2. □

## 7. Problems 24 and 28

Problem 24 of the *Nonarchimedean Scottish Book* asks whether the map of rings underlying a rational localisation of Tate Huber pairs must be flat. Problem 28 asks whether a non-zero-divisor which generates a closed ideal can restrict to zero on a nonempty rational subspace [19, Problems 24 and 28]. The same rational datum, together with multiplicativity of the Gauss norm, answers both questions. We keep the notation of the introduction. In fact, the argument only uses that $k$ is complete and ultrametric and that $0 < |\varpi| < 1$; neither conclusion uses sheafiness. For Problem 28 this answers the Tate-ring case; we do not address the separate perfectoid case.

**Proposition 7.1.** *Let $k$ be any complete ultrametric normed field, choose $\varpi \in k^\times$ with $0 < |\varpi| < 1$, and form $A$ as above. Let*

$$U = \{x \in \mathrm{Spa}(A, A^\circ) : |W(x)| \leq |\varpi(x)|\}.$$

*Then $U$ is nonempty, the rational localisation*

$$A \longrightarrow \mathcal{O}(U) = A\left\langle \frac{W}{\varpi} \right\rangle$$

*is not flat, and multiplication by $Q^2$ is a strict embedding with closed image even though $Q^2$ maps to zero in $\mathcal{O}(U)$.*

*Proof.* The composite

$$A \longrightarrow B = k\langle W, Q\rangle/(Q^2) \longrightarrow k, \qquad f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(0),$$

is norm-nonincreasing. It therefore sends $A^\circ$ into $k^\circ$, and pulling back the norm valuation on $k$ gives a continuous valuation belonging to $\mathrm{Spa}(A, A^\circ)$. At this point $W$ has value zero and $\varpi$ has nonzero value, so the point lies in $U$.

Since the restricted Gauss norm on $A$ is multiplicative and $\|Q\| = 1$, for every $a \in A$ we have

$$\|Q^2a\| = \|a\|.$$

Thus multiplication by $Q^2$ is an isometry. Its map onto its image, with the subspace topology, is consequently a homeomorphism, so the inclusion is strict. Moreover, an isometric image of the complete space $A$ is closed in $A$. Since $\|Q^2\| = 1$, the element $Q^2$ is nonzero; multiplication by it is therefore injective because $A$ is a domain.

On the other hand, the calculation (4) says directly that the canonical map to the completed rational localisation kills $Q^2$. The isomorphism of Proposition 4.1 identifies this localisation with the nonzero ring

$$A \longrightarrow k\langle X, Q\rangle/(Q^2), \qquad W \longmapsto \varpi X, \quad Q \longmapsto Q.$$

If the canonical map were flat, the non-zero-divisor $Q^2$ would still act injectively on $\mathcal{O}(U)$; equivalently, tensoring the injection

$$0 \longrightarrow A \xrightarrow{\cdot Q^2} A$$

with $\mathcal{O}(U)$ would make multiplication by the image of $Q^2$ injective. Its image is zero, however, so this multiplication map is zero on the nonzero ring $\mathcal{O}(U)$, a contradiction. $\square$

## 8. A second example

In the finite-jet construction, the bad rational localisation is non-reduced. In our second example it is an integral domain; the loss of uniformity is instead caused by an unbounded family of non-nilpotent power-bounded elements.

The formal statements in this section use the following base hypotheses: $k$ is a complete nontrivially normed ultrametric field, its norm unit ball is noetherian, and $\varpi$ is a chosen uniformizer (equivalently here, an irreducible element of the norm unit ball). A complete discretely valued field supplies these hypotheses. The domain, uniformity, and non-noetherianity arguments below use only the indicated parts of this package; the noetherianity of the unit ball enters the localisation and sheafiness results.

**Theorem 8.1.** *Let $w : \mathbb{N}_{>0} \to \mathbb{N}$ satisfy $w(n) \geq 1$ for every $n$ and be unbounded. The complete Tate $k$-algebra $\mathcal{A}_w$ defined below has the following properties.*

(1) *$\mathcal{A}_w$ is a uniform, non-noetherian integral domain and $\mathcal{A}_w^\circ = \mathcal{A}_{w,0}$, where $\mathcal{A}_{w,0}$ is defined below.*
(2) *The ring $\mathcal{A}_w$ is strongly sheafy; in particular, the Huber pair $(\mathcal{A}_w, \mathcal{A}_w^\circ)$ is sheafy.*
(3) *The rational localisation*

$$\mathcal{B}_w = \mathcal{A}_w \left\langle \frac{W}{\varpi} \right\rangle$$

*is an integral domain but is not uniform.*

*Consequently $\mathcal{A}_w$ is not stably uniform. In particular, failure of stable uniformity need not be caused by a nilpotent in the bad rational localisation.*

Taking $w(n) = n$ gives the example announced in the introduction. We prove the component statements below for the weakest hypotheses on $w$ that they require.

8.1. **The weighted-parity algebra.** We write $\mathbb{N} = \{0, 1, 2, \ldots\}$, put $I = \mathbb{N}_{>0}$, and fix an arbitrary function $w : I \to \mathbb{N}$. For a finite-support multi-index $\nu = (\nu_n)_{n \in I} \in \mathbb{N}^{(I)}$, put

$$\omega_w(\nu) = \sum_{\substack{n \geq 1 \\ \nu_n \text{ odd}}} w(n). \tag{9}$$

To lighten the notation, write $\omega = \omega_w$ for the rest of the section. Consider the submonoid

$$S = \left\{ (a, \nu) \in \mathbb{N} \times \mathbb{N}^{(I)} : a \geq \omega(\nu) \right\}. \tag{10}$$

The inequality

$$\omega(\nu + \mu) \leq \omega(\nu) + \omega(\mu) \tag{11}$$

shows that $S$ is closed under addition. It is locally finite: a fixed element of $S$ has only finitely many decompositions as a sum of two elements of $S$.

Let $k[S]$ be the monoid algebra with basis $W^a U^\nu$, where $U^\nu = \prod_n U_n^{\nu_n}$. Give it the Gauss norm

$$\left\| \sum_{(a,\nu) \in S} c_{a,\nu} W^a U^\nu \right\|_G = \sup_{a,\nu} |c_{a,\nu}|. \tag{12}$$

Its completion is denoted by $\mathcal{A}_w$, or simply $\mathcal{A}$ when $w$ is understood:

$$\mathcal{A}_w = \left\{ \sum_{(a,\nu) \in S} c_{a,\nu} W^a U^\nu : c_{a,\nu} \in k, \quad c_{a,\nu} \longrightarrow 0 \right\}, \tag{13}$$

where convergence to zero is over the discrete index set $S$. Local finiteness makes the coefficientwise convolution finite, and the submultiplicativity of the Gauss norm on $k[S]$ extends multiplication continuously to $\mathcal{A}$. Put

$$\mathcal{A}_{w,0} = \left\{ \sum_{(a,\nu)\in S} c_{a,\nu} W^a U^\nu \in \mathcal{A}_w : c_{a,\nu} \in k^\circ \right\}.$$

We henceforth abbreviate these rings to $\mathcal{A}$ and $\mathcal{A}_0$. Then $\mathcal{A}_0$ is the closed unit ball for (12), it is $\varpi$-adically complete, and $\mathcal{A} = \mathcal{A}_0[1/\varpi]$.

Put

$$Y_n = W^{w(n)} U_n, \qquad Z_n = U_n^2,$$

and, for $N \in \mathbb{N}$, let $\mathcal{A}^{(N)} \subset \mathcal{A}$ be the closed subring supported on monomials involving only $W, U_1, \dots, U_N$. Thus $\mathcal{A}^{(0)} = k\langle W \rangle$. Let $\mathcal{T}^{(N)} \subset \mathcal{A}^{(N)}$ be the subring in which every $U_n$ has even exponent.

**Lemma 8.2** (Finite-variable subrings)**.** *For every weight $w$ and every $N \in \mathbb{N}$, substitution $Z_n \mapsto U_n^2$ gives a norm-preserving ring isomorphism*

$$k\langle W, Z_1, \dots, Z_N \rangle \xrightarrow{\sim} \mathcal{T}^{(N)}.$$

*For $N = 0$ this is the evident identification of both sides with $k\langle W \rangle$. Every $f \in \mathcal{A}^{(N)}$ can be written*

$$f = \sum_{\epsilon \in \{0,1\}^N} f_\epsilon Y^\epsilon, \qquad f_\epsilon \in \mathcal{T}^{(N)}, \qquad Y^\epsilon = \prod_{n=1}^{N} Y_n^{\epsilon_n}. \tag{14}$$

*Consequently $\mathcal{A}^{(N)}$ is finite over $\mathcal{T}^{(N)}$ and is a strongly noetherian Tate ring. The inclusions $\mathcal{A}^{(N)} \to \mathcal{A}^{(M)} \to \mathcal{A}$ for $N \le M$ preserve the norm. Moreover, for every $f \in \mathcal{A}$ and $\ell \ge 0$ there are $N$ and $f_N \in \mathcal{A}^{(N)}$ such that*

$$\|f - f_N\|_G \le |\varpi|^\ell \|f\|_G.$$

*In particular,*

$$\mathcal{A} = \widehat{\bigcup_{N\ge 0} \mathcal{A}^{(N)}}. \tag{15}$$

*Proof.* On even exponents, halving the $U_n$-coordinates identifies the support monoid with that of the Tate algebra $k\langle W, Z_1, \dots, Z_N \rangle$; the coefficient map and its inverse preserve the Gauss norm. Write each exponent uniquely as $\nu_n = 2q_n + \epsilon_n$, with $\epsilon_n \in \{0, 1\}$. Since $\omega(\nu) = \sum_{\epsilon_n=1} w(n)$, every allowed monomial factors as

$$W^a U^\nu = W^{a-\omega(\nu)} \prod_{n=1}^{N} Z_n^{q_n} \prod_{n=1}^{N} Y_n^{\epsilon_n}.$$

Grouping the monomials with the same tuple $(\epsilon_1, \dots, \epsilon_N)$ gives (14). The finitely many $Y^\epsilon$ therefore generate $\mathcal{A}^{(N)}$ over $\mathcal{T}^{(N)}$. The same argument after adjoining any finite number of Tate variables gives strong noetherianity. The inclusions are coefficientwise and hence isometric. Finally, approximate $f$ to the required precision by a finite sum of monomials and choose $N$ so that only $U_1, \dots, U_N$ occur in that sum. This gives the stated estimate and (15). $\square$

**Proposition 8.3.** *For every weight $w$, the ring $\mathcal{A}_w$ is a complete uniform Tate $k$-algebra and an integral domain, with*

$$\mathcal{A}_w^\circ = \mathcal{A}_{w,0}.$$

*Proof.* The Gauss norm on the countable restricted Tate algebra $k\langle W, U_1, U_2, \dots\rangle$ is multiplicative. To see this, note first that only finitely many coefficients attain the nonzero supremum, since the coefficients tend to zero. Order the finite-support exponent vectors by comparing them at the largest index where they differ. This is an additive total order. For each of two nonzero series choose the largest norm-attaining exponent. In the coefficient of the sum of these two exponents, the product of the chosen coefficients strictly dominates every other convolution term. The ultrametric inequality therefore gives

$$\|fg\|_G = \|f\|_G \|g\|_G.$$

This is the countable restricted-series argument used in the formalisation, whose Lean implementation uses William Coram's restricted-power-series code [6]. The support algebra $\mathcal{A}$ is a closed isometric subring of this Tate algebra, so its Gauss norm is multiplicative and it is a domain.

If $x \in \mathcal{A}_0$, all powers of $x$ remain in $\mathcal{A}_0$. If $x \notin \mathcal{A}_0$, multiplicativity gives $\|x^m\|_G = \|x\|_G^m$, which is unbounded. Hence $\mathcal{A}^\circ = \mathcal{A}_0$. Completeness and the Tate property follow from (13) and the topologically nilpotent unit $\varpi$. □

**Proposition 8.4.** *If $w(n) \geq 1$ for every $n \geq 1$, then $\mathcal{A}_w$ is not noetherian.*

*Proof.* Assume that $w(n) \geq 1$ for every $n \geq 1$. For $m \geq 1$, let

$$I_m = (Z_1, \dots, Z_m) \subset \mathcal{A}.$$

Define a norm-nonincreasing homomorphism

$$\psi_m : \mathcal{A} \longrightarrow k\langle T\rangle,$$

by retaining only the coefficients of the monomials $U_{m+1}^{2q}$ and sending that monomial to $T^q$. The support condition shows directly that this is multiplicative: a product can contribute such a monomial only when both factors contribute monomials of the same form. Thus $\psi_m$ sends $W$ and every $Y_j$ to zero, sends $Z_{m+1}$ to $T$, and sends every other $Z_j$ to zero. It kills $I_m$ but not $Z_{m+1}$. Therefore

$$I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \cdots,$$

so $\mathcal{A}$ is not noetherian. □

## 8.2. **Small perturbations of rational data.**

**Lemma 8.5** (Small perturbation lemma)**.** *Let $(E, E^+)$ be a complete Tate Huber pair and let $\alpha = (f_1, \dots, f_m; g)$ be a rational datum in $E$. There is a neighbourhood $V$ of zero in $E$ such that, whenever*

$$g' - g,\ f_1' - f_1, \dots, f_m' - f_m \in V,$$

*the tuple $\alpha' = (f_1', \dots, f_m'; g')$ is a rational datum and $\alpha$ and $\alpha'$ define the same rational subset of $\mathrm{Spa}(E, E^+)$. Their completed rational localisations are canonically bicontinuously isomorphic over $E$.*

*Proof.* Apply [10, Lemma 3.10] after including $g$ among the numerators; see also [13, Remark 2.4.7]. The assertion about completed rational localisations follows from their universal property [11, Proposition 1.3]. □

8.3. **Rational localisations defined over finitely many variables.** Fix $N \in \mathbb{N}$. Let $\mu = (\mu_n)_{n>N}$ be a finitely supported family of nonnegative integers and put

$$e_\mu = W^{\omega(\mu)} U^\mu,$$

where $\mu$ is extended by zero for $n \le N$. Every $f \in \mathcal{A}$ has a unique convergent expression

$$f = \sum_\mu f_\mu e_\mu, \qquad f_\mu \in \mathcal{A}^{(N)}, \qquad \|f_\mu\|_G \longrightarrow 0, \tag{16}$$

and

$$\|f\|_G = \sup_\mu \|f_\mu\|_G.$$

Multiplication is determined by

$$e_\mu e_\lambda = W^{\omega(\mu)+\omega(\lambda)-\omega(\mu+\lambda)} e_{\mu+\lambda}. \tag{17}$$

The map

$$\rho_N : \mathcal{A} \longrightarrow \mathcal{A}^{(N)}, \qquad \sum_\mu f_\mu e_\mu \longmapsto f_0,$$

is a norm-nonincreasing ring retraction.

**Proposition 8.6** (Coefficientwise localisation)**.** *Let $\alpha$ be a rational datum whose entries lie in $\mathcal{A}^{(N)}$, and put $P = (\mathcal{A}^{(N)})_\alpha$. There is a bicontinuous ring isomorphism*

$$\mathcal{A}_\alpha \xrightarrow{\sim} \left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P, \quad \|p_\mu\| \longrightarrow 0 \right\}.$$

*The ring on the right has the supremum norm, and its multiplication is given by* (17)*, with $W$ replaced by its image in $P$. If a rational refinement is also represented by data in $\mathcal{A}^{(N)}$ and $P \to P'$ is the corresponding restriction map, then the induced restriction sends*

$$\sum_\mu p_\mu e_\mu \longmapsto \sum_\mu (p_\mu|_{P'}) e_\mu.$$

*Proof.* Apply the canonical map $\mathcal{A}^{(N)} \to P$ to every coefficient in (16). This gives a continuous homomorphism

$$\Phi_0 : \mathcal{A} \longrightarrow \left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P, \quad \|p_\mu\| \longrightarrow 0 \right\}.$$

In the target, the image of $g$ is a unit and each $f_i/g$ is of norm at most 1, by the defining property of $P$. Thus $\Phi_0$ extends continuously to $\mathcal{A}[1/g]$ for the rational-localisation topology, and then to the completion:

$$\Phi : \mathcal{A}_\alpha \longrightarrow \left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P, \quad \|p_\mu\| \longrightarrow 0 \right\}.$$

In the other direction, the inclusion $\mathcal{A}^{(N)} \to \mathcal{A}$ gives a continuous homomorphism $P \to \mathcal{A}_\alpha$. Each $e_\mu$ lies in $\mathcal{A}_0$, and the canonical map to a rational localisation is bounded. Hence the family $(e_\mu)_\mu$ is bounded in $\mathcal{A}_\alpha$. Apply the homomorphism to every coefficient. If $\|p_\mu\| \to 0$, the series

$$\sum_\mu p_\mu e_\mu$$

converges in $\mathcal{A}_\alpha$. The resulting map $\Psi$ is continuous, and (17) shows that it is multiplicative.

The composite $\Phi\Psi$ fixes every term $pe_\mu$, and hence every finite sum of such terms. These finite sums are dense. The other composite fixes the dense image of $\mathcal{A}[1/g]$. Continuity therefore gives $\Phi\Psi = 1$ and $\Psi\Phi = 1$. This proves the claimed isomorphism.

If $P \to P'$ comes from a rational refinement, all the maps just used commute with this restriction map. On a finite sum, restriction simply applies $P \to P'$ to each coefficient; continuity gives the stated formula for every convergent sum. □

**Corollary 8.7** (Finite-variable presentation)**.** *For every rational localisation of $\mathcal{A}_w$, there is $N_0$ such that, for every $N \geq N_0$, it is bicontinuously isomorphic to a ring of the form*

$$\left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P, \quad \|p_\mu\| \longrightarrow 0 \right\},$$

*where $P$ is a rational localisation of the strongly noetherian Tate ring $\mathcal{A}^{(N)}$.*

*Proof.* Start with a rational datum $\alpha = (f_1, \dots, f_m; g)$ in $\mathcal{A}$. Choose a scalar $t \in k^\times$ with $0 < |t| < 1$ and small enough that $tg, tf_1, \dots, tf_m$ lie in $\mathcal{A}_0$. Multiplying the datum by $t$ does not change the rational subset or its localisation, so we may assume that $g, f_1, \dots, f_m$ lie in $\mathcal{A}_0$. Choose a neighbourhood $V$ as in Lemma 8.5. By (15), for some $N$ there are $g', f'_1, \dots, f'_m \in \mathcal{A}^{(N)} \cap \mathcal{A}_0$ such that

$$g' - g,\ f'_1 - f_1, \dots, f'_m - f_m \in V.$$

By Lemma 8.5, the datum $\alpha' = (f'_1, \dots, f'_m; g')$ gives the same rational subset and the same completed localisation. It is rational in $\mathcal{A}$, so applying $\rho_N$ to a Bezout identity for its entries shows that it is already a rational datum in $\mathcal{A}^{(N)}$. The same datum belongs to every larger finite-variable ring, and the corresponding retraction shows from the same Bezout identity that it is rational there. Now apply Proposition 8.6 at every stage beyond $N$. □

8.4. **A reduced rational chart which is not uniform.** The ideal $(W, \varpi)$ is open because $\varpi$ is a unit of $\mathcal{A}$. Define the complete $k^\circ$-algebra

$$\mathcal{B}_{w,0} = \left\{ \sum_{\nu \in \mathbb{N}^{(I)},\ d \geq \omega(\nu)} b_{d,\nu} X^d U^\nu : b_{d,\nu} \in \varpi^{\omega(\nu)} k^\circ, \quad \varpi^{-\omega(\nu)} b_{d,\nu} \longrightarrow 0 \right\}, \tag{18}$$

with norm

$$\left\| \sum b_{d,\nu} X^d U^\nu \right\|_{\mathcal{B}} = \sup_{d,\nu} |\varpi|^{-\omega(\nu)} |b_{d,\nu}|. \tag{19}$$

The subadditivity (11) shows that $\mathcal{B}_{w,0}$ is a ring. Put $\mathcal{B}_w = \mathcal{B}_{w,0}[1/\varpi]$, and abbreviate these rings to $\mathcal{B}$ and $\mathcal{B}_0$ when $w$ is understood.

**Proposition 8.8.** *For every weight $w$, there is a bicontinuous ring isomorphism*

$$\mathcal{A}_w \left\langle \frac{W}{\varpi} \right\rangle \xrightarrow{\sim} \mathcal{B}_w, \qquad W \longmapsto \varpi X.$$

*Proof.* The datum $(W; \varpi)$ lies in $\mathcal{A}^{(0)} = k\langle W \rangle$. Put

$$P = \mathcal{A}^{(0)}\langle T \rangle / \overline{(\varpi T - W)}.$$

Evaluation at $W = \varpi X$ and $T = X$ gives a continuous map $P \to k\langle X \rangle$. Conversely, evaluation of $k\langle X \rangle$ at the image of $T$ gives a continuous map in the other direction. The two maps are inverse by polynomial density, and the norm estimates in the two universal properties show that the isomorphism is isometric.

Apply Proposition 8.6 with $N = 0$. It describes the rational chart as the ring of sums

$$\sum_{\nu} p_\nu e_\nu, \qquad p_\nu \in P, \qquad \|p_\nu\| \longrightarrow 0.$$

Under $P \cong k\langle X\rangle$, this is precisely the model (18). Its sup norm is (19), because the term indexed by $\nu$ is multiplied by $\varpi^{\omega(\nu)}$. The map from $\mathcal{A}^{(0)}$ sends $W$ to $\varpi X$. □

**Proposition 8.9.** *For every weight $w$, the ring $\mathcal{B}_w$ is an integral domain. If $w$ is unbounded, then $\mathcal{B}_w$ is not uniform.*

*Proof.* Put $R = k\langle X\rangle$. Here

$$R[[U_1, U_2, \ldots]] = \left\{ \sum_{\nu \in \mathbb{N}^{(I)}} a_\nu U^\nu : a_\nu \in R \right\}$$

denotes the ring of all coefficient families indexed by finite-support multi-indices. Multiplication is convolution. For a fixed $\nu$ there are only finitely many decompositions $\nu = \lambda + \mu$, so every coefficient of a product is a finite sum. Order $\mathbb{N}^{(I)}$ by comparing two vectors at the largest index where they differ. This is an additive well-order. If two series are nonzero, the coefficient at the sum of their least nonzero exponents is the product of the corresponding coefficients and is nonzero. Thus $R[[U_1, U_2, \ldots]]$ is a domain.

The coefficient description gives a homomorphism

$$\mathcal{B} \longrightarrow R[[U_1, U_2, \ldots]], \qquad \sum_{\nu} p_\nu e_\nu \longmapsto \sum_{\nu} (\varpi X)^{\omega(\nu)} p_\nu(X) U^\nu.$$

It is a homomorphism by (17). If its image is zero, then

$$(\varpi X)^{\omega(\nu)} p_\nu(X) = 0$$

for every $\nu$. Since $k\langle X\rangle$ is a domain, every $p_\nu$ is zero. The map is therefore injective, and hence $\mathcal{B}$ is a domain.

Assume now that $w$ is unbounded. Inside $\mathcal{A}$ write $Y_n = W^{w(n)} U_n$ and put

$$T_n = \varpi^{-w(n)} Y_n = X^{w(n)} U_n \in \mathcal{B}.$$

Formula (19) gives

$$\|T_n\|_{\mathcal{B}} = |\varpi|^{-w(n)}, \qquad \|T_n^2\|_{\mathcal{B}} = 1.$$

The even powers of $T_n$ have norm at most 1, and the odd powers have norm at most $\|T_n\|_{\mathcal{B}}$. Thus every $T_n$ is power-bounded. The family $(T_n)_{n \geq 1}$ is not bounded, since its norms $|\varpi|^{-w(n)}$ are unbounded. Hence $\mathcal{B}^\circ$ is unbounded and $\mathcal{B}$ is not uniform. □

## 8.5. **Strong sheafiness.**

**Theorem 8.10.** *For every weight $w$, every $s \geq 0$, and every ring of integral elements $\mathcal{A}^+_{w,s}$ of $\mathcal{A}_w\langle V_1, \ldots, V_s\rangle$, the Huber pair*

$$\big(\mathcal{A}_w\langle V_1, \ldots, V_s\rangle, \mathcal{A}^+_{w,s}\big)$$

*is sheafy. In particular, $\mathcal{A}_w$ is strongly sheafy. No positivity or unboundedness hypothesis on $w$ is needed here.*

*Proof.* We first work with a finite rational cover $U = \bigcup_i U_i$. Choose one $N$ large enough to approximate the datum for $U$ and the data for all the $U_i$ in $\mathcal{A}^{(N)}$. The standard combined data for every pairwise intersection $U_i \cap U_j$ then also lie in $\mathcal{A}^{(N)}$. The small perturbation lemma does not change the rational domains, and Proposition 8.6 gives compatible identifications

$$\mathcal{O}(U) \cong \left\{ \sum_\mu p_\mu e_\mu : p_\mu \in P,\ \|p_\mu\| \to 0 \right\},$$

$$\mathcal{O}(U_i) \cong \left\{ \sum_\mu p_{i,\mu} e_\mu : p_{i,\mu} \in P_i,\ \|p_{i,\mu}\| \to 0 \right\},$$

with coefficientwise restriction maps. Here $P$ is a rational localisation of the strongly noetherian Tate ring $\mathcal{A}^{(N)}$, and the $P_i$ are rational localisations corresponding to the $U_i$. Let $V$ and $V_i$ be the rational subsets of

$$\mathrm{Spa}\big(\mathcal{A}^{(N)}, (\mathcal{A}^{(N)})^\circ\big)$$

defined by the data giving $P$ and $P_i$. Then $V = \bigcup_i V_i$. Indeed, if $v \in V$, the composite $v \circ \rho_N$ is a point of $\mathrm{Spa}(\mathcal{A}, \mathcal{A}^\circ)$, since $\rho_N$ sends power-bounded elements to power-bounded elements. For data in $\mathcal{A}^{(N)}$, membership in the corresponding rational subset is unchanged by this pullback because $\rho_N$ is a retraction. Thus $v \circ \rho_N$ belongs to $U$, and hence to some $U_i$, which gives $v \in V_i$. We may therefore apply sheafiness of the strongly noetherian ring $\mathcal{A}^{(N)}$ to the cover $V = \bigcup_i V_i$.

Let $(x_i)_i$ be a matching family and write $x_i = \sum_\mu x_{i,\mu} e_\mu$. Since $\mathcal{A}^{(N)}$ is sheafy, the matching family $(x_{i,\mu})_i$ glues to a unique $q_\mu \in P$ for every $\mu$. It remains to check that $\|q_\mu\| \to 0$. The restrictions of the $q_\mu$ to every $P_i$ tend to zero because the coefficients of each $x_i$ do. The product restriction map

$$P \longrightarrow \prod_i P_i$$

is a topological embedding, so it reflects convergence to zero. Hence $q_\mu \to 0$, and $\sum_\mu q_\mu e_\mu$ is the required section over $U$. The same coefficientwise argument proves separation.

The image of the product restriction map for $\mathcal{A}$ is therefore the subspace of tuples whose restrictions to every pairwise intersection agree. This subspace is closed: it is the common zero locus of the continuous difference maps, and hence is a Banach $k$-space. The restriction map is a continuous bijection from the Banach $k$-space $\mathcal{O}(U)$ to this closed subspace. The nonarchimedean open mapping theorem therefore makes it a homeomorphism. We have proved the topological sheaf condition on finite rational covers. The general rational-basis theorem upgrades this to the all-open structure presheaf, and its plus-ring independence gives the same conclusion for every ring of integral elements. In particular, $(\mathcal{A}, \mathcal{A}^\circ)$ is sheafy.

For $s \geq 0$, the Tate extension $\mathcal{A}\langle V_1, \ldots, V_s\rangle$ is bicontinuously isomorphic to the same weighted-parity construction with the first $s$ weights equal to zero and the remaining weights shifted by $s$. The argument above applies to every weight function and is independent of the ring of integral elements, so this Tate extension is sheafy for every such choice. Hence $\mathcal{A}$ is strongly sheafy. $\square$

*Proof of Theorem 8.1.* The fact that $\mathcal{A}$ is a uniform domain is Proposition 8.3, and its failure to be noetherian is Proposition 8.4. Strong sheafiness is Theorem 8.10. Finally, Propositions 8.8 and 8.9 identify the rational chart $|W| \leq |\varpi|$ as an integral domain which is not uniform. $\square$

**Remark 8.11** (Admissible blow-ups)**.** There is a common formal interpretation of the two bad rational charts. Let $R_0$ denote $A_0$ in the finite-jet example and $\mathcal{A}_0$ in the weighted-parity example. The affine chart of the admissible blow-up of $\mathrm{Spf}(R_0)$ along $(\varpi, W)$ on which the pullback of this ideal is generated by $\varpi$ has ring

$$\widehat{R_0[W/\varpi]}.$$

Here $R_0[W/\varpi]$ means the subring of $R_0[1/\varpi]$ generated by $R_0$ and $W/\varpi$, and the hat denotes the $\varpi$-adic completion defined above. After inverting $\varpi$, its coordinate ring is the completed rational localisation $R_0[1/\varpi]\langle W/\varpi\rangle$, corresponding to $|W| \leq |\varpi|$. This description applies to our non-noetherian formal models because they are $\varpi$-adic and their ideal of definition $(\varpi)$ is principal; see [7, Chapter 0, Section 7.1(c), and Chapter II, Section 1.1(b)].

For $R_0 = A_0$, the ring $A_0[W/\varpi]$ is a domain, but the calculation in the proof of Proposition 4.1 shows that

$$Q^2 \in \bigcap_{n \geq 1} \varpi^n A_0[W/\varpi].$$

This intersection is the kernel of the completion map. Thus $Q^2$ is killed, while $Q$ survives, so this rational chart is non-reduced. The map from $A$ to this completed rational localisation is also not flat by Proposition 7.1. For $R_0 = \mathcal{A}_0$, the completed chart remains a domain, but its generic fibre is not uniform because the power-bounded elements $T_n = X^{w(n)}U_n$ form an unbounded family. Thus the two examples give different failures on the same kind of affine blow-up chart.

The proof of sheafiness for $\mathcal{A}$ uses the further fact that a rational datum can be approximated inside some strongly noetherian $\mathcal{A}^{(N)}$, after which localisation and restriction are computed coefficient by coefficient. The infinite monomial support and the unbounded power-bounded elements are in the tradition of the examples of Buzzard–Verberkmoes and Mihara; the approximation by the rings $\mathcal{A}^{(N)}$ is what makes sheafiness accessible here.

## 9. How the examples were found

The authors had for some time been interested in finding a counterexample to Problem 7 of the *Nonarchimedean Scottish Book*. Our attempts had centred on the example of [4, Section 4.5, Proposition 17], which is uniform but not stably uniform. We believe that this example is in fact sheafy, but we have not found a proof.

In January 2026 we began to experiment with AI tools, initially ChatGPT 5.2 Pro. Buzzard–Verberkmoes reduce exactness for a two-member Laurent cover to a boundedness condition on the relevant rings of definition [4, Lemma 2]. For a uniform ring this follows from boundedness of the power-bounded elements [4, Lemma 3 and Corollary 4]. Their reduction from general rational covers to Laurent covers is given in [4, Lemma 8 and the proof of Theorem 7]. For sheafiness these checks must also be made after rational localisation. Their Theorem 7 cannot be applied directly to Proposition 17, since one of the rational localisations in that example is not uniform. Our plan was instead to ask ChatGPT to write down many explicit rational covers and check the required exactness directly for each one, in the hope that a pattern, and eventually a proof, would emerge.

After a couple of weeks ChatGPT 5.2 Pro claimed to have a complete proof. The authors then began the slow process of understanding this proof, only to discover, with the help of ChatGPT 5.3, that one of its arguments used the following assertion:

*If $\phi : A \to B$ is a continuous homomorphism of Hausdorff topological abelian groups and $A$ is complete, then $\phi(A)$ is closed in $B$.*

This assertion is false, and its use completely broke the purported proof.

Motivated by this, we began in parallel to use Claude Code to formalise the parts of the theory of adic spaces needed here, including the theorem that strongly noetherian Tate rings are sheafy. The aim was a practical one: future AI-generated proofs could be formalised before the human authors had spent days understanding an argument containing a fatal error near its end.

We then returned to writing down covers and looking for a pattern. This continued through ChatGPT 5.3 and subsequent versions. When ChatGPT 5.6 Sol became available, we asked, almost in passing, whether it could prove the result we wanted or *find a new example*. It produced the examples studied here. By then the Lean development was far enough advanced that we could use Claude Code to formalise the arguments quickly in Lean and check that they were correct.

## Appendix A. Lean formalisation and verification

This appendix records the Lean formalisation of the two examples. The core finite-jet declarations are taken from AINTLIB commit `b007a4f3d`, and the general-base finite-jet and Scottish Book declarations from commit `01116aca6`. The weighted-parity declarations are taken from commit `090a28921`, and the comparison with Mathlib's sheaf predicate from commit `d92f96504`. The generic Koszul, pullback, Milnor-descent, and strong-sheafiness declarations are taken from commit `870d0eed2` [1]. For each example Lean proves the Tate structure and uniformity, proves that the ring is a domain but not noetherian, proves sheafiness and failure of stable uniformity, and verifies the rational chart calculation used in the proof.

All five pinned commits are public in AINTLIB. The interactive version of the paper includes each declaration used below, extracted from the appropriate commit, and its archival page links to the pinned source on GitHub. The verification report records the Lean and mathlib versions, the build, and the axiom checks.

The code excerpts below retain the mathematically meaningful bodies of definitions. Proof fields which only verify closure are omitted. For theorems we display only the statement and replace the proof by `...`; the links give the complete Lean declarations and their checked proofs.

### A.1. **The rational-cover predicate `IsSheafy`.**

**Definition A.1** (The rational-cover predicate)**.** The proofs are carried out on finite rational covers, so the formalisation uses the class `IsSheafy`. We chose this formulation because it is easier to work with for our purposes: it speaks directly about the completed rational localisations which occur in the proofs. The equivalences below show that this choice does not change the notion of sheafiness. The class is in the `ValuationSpectrum` namespace and is defined in `StructureSheaf.lean`. Here `RationalCoveringData A` consists of a rational datum for the base and finitely many data whose rational subsets cover it. The separate condition `C.IsRational` says that the numerator set in the base and in each member of the cover generates an open ideal. For a Tate ring this is equivalent to generating the unit ideal. Finally, `presheafValue D` is the completed rational localisation $A\langle T/s\rangle$ attached to $D$.

```
class IsSheafy (A : Type u) [CommRing A] [TopologicalSpace A]
    [IsTopologicalRing A] [inst₁ : PlusSubring A]
    [inst₂ : IsHuberRing A] [T2Space A] [NonarchimedeanRing A]
    [letI : UniformSpace A :=
      IsTopologicalAddGroup.rightUniformSpace A; CompleteSpace A]
    [IsRingOfIntegralElements (A⁺)] : Prop where
  embedding : ∀ (C : RationalCoveringData A), C.IsRational →
    Topology.IsEmbedding (productRestrictionSub A C)
  gluing : ∀ (C : RationalCoveringData A), C.IsRational →
    ∀ (f : ∀ (D : ↑C.covers), presheafValue D.1),
    (∀ (D₁ D₂ : ↑C.covers) (D₃ : RationalLocData A)
      (h₃₁ : rationalOpen D₃.T D₃.s ⊆
        rationalOpen D₁.1.T D₁.1.s)
      (h₃₂ : rationalOpen D₃.T D₃.s ⊆
        rationalOpen D₂.1.T D₂.1.s),
```

```
      restrictionMap D₁.1 D₃ h₃₁ (f D₁) =
        restrictionMap D₂.1 D₃ h₃₂ (f D₂)) →
    ∃ x : presheafValue C.base, ∀ (D : ↑C.covers),
      restrictionMap C.base D.1 (C.hsubset D.1 D.2) x = f D
```

The two fields say that restriction to a finite rational cover is a topological embedding and that every matching family glues. Compatibility is written using common rational refinements; the exact-intersection theorem identifies this with the usual compatibility on pairwise intersections. Thus `IsSheafy A` is the topological sheaf condition on the rational basis.

## A.2. **Comparison with Mathlib's sheaf predicate.**

**Definition A.2** (Sections on arbitrary opens)**.** Let $V \subseteq \mathrm{Spa}(A, A^+)$ be open. The ring `limitSections V` consists of compatible sections over the rational subdomains contained in $V$, or equivalently their projective limit. If $W \subseteq V$, then `limitRestrict` is the map $\mathcal{O}(V) \to \mathcal{O}(W)$ obtained by reindexing this family. The corresponding all-open condition is `IsLimitSheaf`, from `SheafyPair.lean`.

The class `HasLocLiftPowerBounded` is used internally to construct the restriction maps between completed rational localisations. For an inclusion $R(D') \subseteq R(D)$, it says that the denominator of $D$ becomes a unit on $R(D')$ and that the fractions $t/s$, for $t$ in the numerator set of $D$, are power-bounded there. This class is supplied automatically for the complete Tate pairs considered here; it is not an additional hypothesis in our results.

```
structure IsLimitSheaf (A : Type u) [CommRing A] [TopologicalSpace A]
    [IsTopologicalRing A] [PlusSubring A] [IsHuberRing A]
    [HasLocLiftPowerBounded A] : Prop where
  injective : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺)))
    {x y : ↑(limitSections V)},
    (∀ i, limitRestrict (hle i) x = limitRestrict (hle i) y) → x = y
  glue : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺)))
    (s : ∀ i, ↑(limitSections (U i))),
    (∀ i j : ι,
      limitRestrict (inf_le_left (a := U i) (b := U j)) (s i) =
      limitRestrict (inf_le_right (a := U i) (b := U j)) (s j)) →
    ∃ x : ↑(limitSections V), ∀ i, limitRestrict (hle i) x = s i
  isEmbedding : ∀ {V : Opens ↑(Spa A A⁺)} {ι : Type u}
    {U : ι → Opens ↑(Spa A A⁺)}
    (hle : ∀ i, U i ≤ V)
    (_ : (V : Set ↑(Spa A A⁺)) ⊆ ⋃ i, (U i : Set ↑(Spa A A⁺))),
    Topology.IsEmbedding (limitRestrictProd hle)
```

The fields `injective` and `glue` are the usual sheaf axioms on arbitrary opens. The final field says that restriction identifies the sections over $V$ homeomorphically with the subspace of $\prod_i \mathcal{O}(U_i)$ consisting of tuples whose restrictions agree. Together these conditions give the concrete form of Wedhorn's sheaf condition for topological rings.

**Remark A.3** (Comparison with Mathlib)**.** For complete Tate pairs the rational-basis and all-open predicates are equivalent:

```
theorem isSheafy_iff_isLimitSheaf :
    IsSheafy A ↔ IsLimitSheaf A := by
  ...
```

The forward implication is the sheaf-on-a-rational-basis theorem, and the converse is restriction to rational covers. For a bundled presheaf $F$, `F.IsSheafOfTopologicalRings` means that, for every topological commutative ring $T$, the presheaf $U \mapsto \mathrm{Hom}_{\mathrm{cont}}(T, F(U))$ is a sheaf of sets. This is Wedhorn's representable formulation [21, Remark 8.20]:

```
def IsSheafOfTopologicalRings : Prop :=
  ∀ (T : Type u) (_ : CommRing T) (_ : TopologicalSpace T)
    (_ : IsTopologicalRing T), ∀ {ι : Type u} (U : ι → Opens X)
    (f : ∀ i, {g : T →+* (F.obj (op (U i)) : Type u) //
      Continuous g}),
    (∀ i j : ι,
      (F.map (homOfLE (inf_le_left (a := U i) (b := U j))).op).1.comp
          (f i).1 =
        (F.map (homOfLE (inf_le_right (a := U i) (b := U j))).op).1.comp
          (f j).1) →
    ∃! g : {g : T →+* (F.obj (op (iSup U)) : Type u) //
      Continuous g},
      ∀ i, (F.map (homOfLE (le_iSup U i)).op).1.comp g.1 = (f i).1
```

For the structure presheaf the formalisation proves:

```
theorem structurePresheaf_isSheafOfTopologicalRings_iff :
    TopCat.Presheaf.IsSheafOfTopologicalRings
        (structurePresheaf A) ↔
      IsLimitSheaf A := by
  ...
```

The preceding theorem, together with `isSheafy_iff_isLimitSheaf`, gives

$$\texttt{IsSheafy A} \iff \texttt{(structurePresheaf A).IsSheafOfTopologicalRings}.$$

Thus the bundled `IsSheafOfTopologicalRings` condition and the rational-cover condition used in the proofs are equivalent; neither is merely a consequence of the other.

There is also a direct comparison with Mathlib's categorical definition. For an arbitrary presheaf $F$ of complete separated topological commutative rings, the general comparison theorem is

```
theorem isSheafOfTopologicalRings_iff_forgetToTopCommRingCat_isSheaf :
    F.IsSheafOfTopologicalRings ↔
      TopCat.Presheaf.IsSheaf
        (F ⋙ CompleteTopCommRingCat.forgetToTopCommRingCat) := by
  ...
```

The functor in this statement forgets completeness and separatedness, but not the topology. The objects of `TopCommRingCat` are all topological commutative rings, so Mathlib's categorical condition tests exactly the presheaves $U \mapsto \mathrm{Hom}_{\mathrm{cont}}(T, F(U))$ appearing in Wedhorn's definition. By contrast, the categorical condition with values left in `CompleteTopCommRingCat` tests only complete separated rings and is only a consequence of Wedhorn's condition.

Specialising the comparison to the structure presheaf gives the all-open equivalence. Its statement is

```
theorem structurePresheaf_forgetToTopCommRingCat_isSheaf_iff :
    TopCat.Presheaf.IsSheaf
        (structurePresheaf A ≫
          CompleteTopCommRingCat.forgetToTopCommRingCat) ↔
      IsLimitSheaf A := by
  ...
```

Combining this with the rational-basis theorem above gives the final comparison:

```
theorem isSheafy_iff_structurePresheaf_forgetToTopCommRingCat_isSheaf
    [DecidableEq A] [DecidableEq (RationalLocData A)] [IsTateRing A]
    [IsRingOfIntegralElements (A⁺ : Subring A)] [T2Space A]
    [NonarchimedeanRing A]
    [letI : UniformSpace A :=
      IsTopologicalAddGroup.rightUniformSpace A; CompleteSpace A] :
    IsSheafy A ↔
      TopCat.Presheaf.IsSheaf
        (structurePresheaf A ≫
          CompleteTopCommRingCat.forgetToTopCommRingCat) := by
  ...
```

Thus the rational-cover predicate, the all-open predicate, Wedhorn's Hom-sheaf condition, and Mathlib's categorical predicate are equivalent. We use `IsSheafy` because it is the convenient form for the arguments in this paper, not because it defines a weaker or stronger notion of sheafiness.

A.3. **Choice-independent sheafiness for Tate rings.** For a complete Tate ring the following definitions fix a ring of integral elements and then remove the choices of plus ring and completion. They are in `SheafyRing.lean`.

```
def IsSheafyFor (Aplus : RingOfIntegralElements A) : Prop :=
  letI := Aplus.toPlusSubring
  haveI : IsRingOfIntegralElements (A⁺ : Subring A) := Aplus.2
  haveI : HasLocLiftPowerBounded A := hasLocLiftPowerBounded_faithful
  IsLimitSheaf A

def IsSheafyComplete : Prop :=
  ∀ Aplus : RingOfIntegralElements A, IsSheafyFor A Aplus

def IsSheafyTateRing : Prop :=
  ∀ (P : PairOfDefinition A)
    (Bplus : RingOfIntegralElements (CompletionModel A P)),
    IsSheafyFor (CompletionModel A P) Bplus
```

The predicate `IsSheafyFor A Aplus` says that the Huber pair $(A, A^+)$ is sheafy. For complete Tate rings, the chosen-plus-ring equivalence proves that the choice of integral elements does not affect the condition. The equivalences in `SheafyCompletionModel.lean` remove the choice of completion. Consequently `IsSheafyTateRing` is the analytic specialisation of Wedhorn's Definition 8.26 [21, Definition 8.26].

A.4. **Strong noetherianity and sheafiness.**

**Definition A.4** (Strong noetherianity)**.** We use the following definitions. They are from `RestrictedPowerSeries.lean`.

```
def MvPowerSeries.IsRestricted {k : ℕ} {A : Type*}
    [CommRing A] [TopologicalSpace A]
    (f : MvPowerSeries (Fin k) A) : Prop :=
  Tendsto (fun s : Fin k →₀ ℕ => MvPowerSeries.coeff s f)
    cofinite (nhds 0)

class IsStronglyNoetherian (A : Type*) [CommRing A]
    [TopologicalSpace A] [NonarchimedeanRing A] : Prop where
  isNoetherianRing_restricted : ∀ k : ℕ,
    IsNoetherianRing (restrictedMvPowerSeriesSubring k A)
```

The cofinite convergence condition is the usual condition that the coefficients tend to zero, so `restrictedMvPowerSeriesSubring k A` represents $A\langle T_1, \ldots, T_k\rangle$. The second declaration therefore says that these rings are noetherian for every $k$. Since $A$ is already complete, this is Wedhorn's definition of strong noetherianity [21, Proposition and Definition 6.36].

For a Tate ring which is not assumed complete, Lean packages the definition using one completion model. The completion-comparison theorem proves that one model is equivalent to every model. Wedhorn's sheafiness theorem is then formalised at the same level of generality:

```
def IsStronglyNoetherianTateRing [IsTateRing A] : Prop :=
  ∃ P : PairOfDefinition A,
    IsStronglyNoetherian (CompletionModel A P)

theorem isSheafyTateRing_of_stronglyNoetherianTateRing
    [IsTateRing A] (h : IsStronglyNoetherianTateRing A) :
    IsSheafyTateRing A := by
  ...
```

Thus one strongly noetherian completion model implies sheafiness for every completion model and every ring of integral elements. For a complete Tate ring the specialization is:

```
theorem isSheafyFor_of_stronglyNoetherianTate
    [IsStronglyNoetherian A]
    (Aplus : RingOfIntegralElements A) :
    IsSheafyFor A Aplus := by
  ...

theorem isSheafyComplete_of_stronglyNoetherianTate
    [IsStronglyNoetherian A] :
    IsSheafyComplete A := by
  ...
```

These are the formal versions of Wedhorn's theorem that a strongly noetherian Tate ring is sheafy [21, Theorem 8.28(b)].

### A.5. **The finite-jet example.**

**Definition A.5** (The finite-jet ring)**.** We finally record the definition of the example itself. In the following code, `PowerSeries.Restricted R 1` is the radius-one Tate algebra in one variable over $R$, `RestrictedLaurent K` is $K\langle W, W^{-1}\rangle$, and `DualNumber R` is $R[Q]/(Q^2)$. The source is `FJP/Over/JetRings.lean`.

```
abbrev L : Type u := RestrictedLaurent K

abbrev JetC : Type u := PowerSeries.Restricted (L K) (1 : ℝ)
```

```
abbrev JetB : Type u :=
  DualNumber (PowerSeries.Restricted K (1 : ℝ))

abbrev JetD : Type u := DualNumber (L K)

noncomputable def jetSupport : Subring (JetC K) where
  carrier := {f |
    qCoeff K 0 f ∈ nonnegSubring K ∧
    qCoeff K 1 f ∈ nonnegSubring K}
  zero_mem' := by
    ...
  one_mem' := by
    ...
  add_mem' := by
    ...
  neg_mem' := by
    ...
  mul_mem' := by
    ...

abbrev JetA : Type u := ↑(jetSupport K)
```

The predicate `nonnegSubring K` cuts out $K\langle W\rangle$ inside $K\langle W, W^{-1}\rangle$. Consequently the carrier condition says that the coefficients of $Q^0$ and $Q^1$ have no negative powers of $W$, while the coefficients of $Q^n$ for $n \geq 2$ are unrestricted. Thus `JetA K` is

$$\left\{ f_0(W) + Qf_1(W) + Q^2 h(W, W^{-1}, Q) \;\middle|\; \substack{f_0, f_1 \in K\langle W\rangle \\ h \in L\langle Q\rangle} \right\},$$

which is the coefficient description of $A$ in (2) after inverting $\varpi$.

The file also constructs the four maps in the Milnor square and proves `milnorRow_exact`: every compatible pair in `JetB K` and `JetC K` comes from a unique element of `JetA K`. This identifies `JetA K` with $B \times_D C$. The sheaf argument produces the finite-rational-cover condition in the following theorem:

```
include ϖ hK₀ in
theorem isSheafy_JetA :
    ValuationSpectrum.IsSheafy (JetA K) := by
  ...
```

Here $\varpi$ is a `Uniformizer K`, and the remaining input is the noetherianity of `unitBall K`. This is the version of sheafiness proved directly for the example. The equivalences above give the all-open and choice-independent statements. The main theorems are collected in `SheafyEndpoints`, and include the following sheafiness statements.

```
theorem finiteJet_isSheafyFor_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    IsSheafyFor (JetA K) (finiteJetPlus K) := by
  ...

theorem finiteJet_isSheafyTateRing_of_dvr
    [IsDiscreteValuationRing 𝒪[K]] :
    IsSheafyTateRing (JetA K) := by
  ...
```

Here $K$ is any complete nontrivially normed ultrametric field for which $\mathcal{O}[K] = \{x : \|x\| \leq 1\}$ is a discrete valuation ring. No local compactness, finiteness of the residue field, characteristic, or perfection assumption is used. The theorem `finiteJet_isSheafyFor_of_dvr` proves that $(A, A^\circ)$ is sheafy, whilst `finiteJet_isSheafyTateRing_of_dvr` removes the choices of completion and ring of integral elements. The corresponding all-open statement is

```
TopCat.Presheaf.IsSheafOfTopologicalRings
    (ValuationSpectrum.structurePresheaf (JetA K)),
```

The corresponding Lean theorem proves this statement directly.

Strong sheafiness is expressed by the following theorem from `FJP/StrongSheafy.lean`.

```
variable (F : Type*) [Field F]

theorem finiteJet_tateExt_isSheafyComplete (n : ℕ) :
    letI := mvTateAlgebraTopology' (A := JetA F) n
    haveI := mvTate_isTateRing (A := JetA F) n
    haveI := mvTate_t2Space (A := JetA F) n
    haveI := mvTate_nonarchimedean (A := JetA F) n
    haveI := mvTateAlgebraTopology'_isTopologicalRing
      (A := JetA F) n
    haveI : @CompleteSpace
      ↑(restrictedMvPowerSeriesSubring n (JetA F))
      (IsTopologicalAddGroup.rightUniformSpace _) :=
        finiteJet_tateExt_completeSpace F n
    IsSheafyComplete
      ↑(restrictedMvPowerSeriesSubring n (JetA F)) := by
  ...
```

The restricted power-series subring in this statement is $A\langle V_1, \ldots, V_n\rangle$, with its canonical Tate-algebra topology, and `IsSheafyComplete` says that it is sheafy for every ring of integral elements. This strong-sheafiness endpoint is currently formalised for the Laurent-series model over $F((t))$. The ordinary sheafiness, uniformity, domain and chart theorems above have the more general DVR-base scope used elsewhere in the appendix.

A.6. **Scope of the finite-jet formalisation.** The paper's $A_0$ is represented by the Gauss-norm unit ball. The theorem `isPowerBounded_JetA_iff` identifies this unit ball with the power-bounded elements, giving $A^\circ = A_0$. Completeness and the Tate structure are supplied by the corresponding instances. As in the weighted-parity development, the constant-series map from $K$ is explicit and isometric but is not installed as an `Algebra K` instance.

For the bad chart, `chartEquiv` is an exported ring equivalence. The two continuity theorems make it a homeomorphism, and the canonical-map theorem identifies $W$ and $Q$. It is not bundled as a `K-AlgEquiv`, which is why Proposition 4.1 is stated for complete topological rings.

The results in section 7 are formalised for any complete ultrametric normed field $K$ equipped with a nonzero element $\varpi$ satisfying $\|\varpi\| < 1$; this is the class `IsFJPBase K`.

```
class IsFJPBase (K : Type u) [NormedField K]
    [IsUltrametricDist K] [CompleteSpace K] where
  pseudoUniformizer : K
  pseudoUniformizer_ne_zero : pseudoUniformizer ≠ 0
  norm_pseudoUniformizer_lt_one :
    ‖pseudoUniformizer‖ < 1
```

At this weaker level Lean proves the conclusions of Propositions 3.1, 4.1 and 4.2; discreteness is not used:

```
variable (K : Type*) [NormedField K] [IsUltrametricDist K]
  [CompleteSpace K] [IsFJPBase K]

theorem finiteJet_isUniform :
    TopologicalRing.IsUniform (JetA K) := by
  ...

theorem finiteJet_isDomain : IsDomain (JetA K) := by
  ...

theorem finiteJet_not_noetherian :
    ¬ IsNoetherianRing (JetA K) := by
  ...

theorem finiteJet_not_stablyUniform :
    ¬ TopologicalRing.IsStablyUniform (JetA K) := by
  ...
```

In fact, the domain and non-noetherianity declarations do not use `IsFJPBase`; the pseudouniformizer is needed for the Tate topology, uniformity, and the bad chart.

Sheafiness is a separate assertion in this newer interface. The corresponding Lean theorem assumes the additional class `IsFJPNoetherianBase K`, which supplies the noetherian Gauss-unit-ball input used in the sheaf proof. The independent `FiniteJetOver` development used for the main theorem proves sheafiness under the complete discretely valued hypothesis of the paper.

Under the weaker `IsFJPBase` hypothesis, the theorem `finiteJet_problem28` records in one statement that $(W; \varpi)$ is rational, its rational subset and section ring are nonempty and nonzero, multiplication by $Q^2$ is a continuous strict injection with closed image, and $Q^2$ restricts to zero. The nonempty point is constructed from the norm valuation pulled back along `ev00`, the map

$$f_0(W) + Qf_1(W) + Q^2h \longmapsto f_0(0).$$

The separate declarations saying that multiplication by $Q^2$ is an isometry and that $Q^2$ restricts to zero are the inputs used below.

```
variable (F : Type*) [NormedField F] [IsUltrametricDist F]
  [CompleteSpace F] [IsFJPBase F]

theorem finiteJet_problem28 :
    (chartDatum F).IsRational ∧
    (rationalOpen (chartDatum F).T (chartDatum F).s).Nonempty ∧
    Nontrivial (presheafValue (chartDatum F)) ∧
    Function.Injective
      (fun a : JetA F => scottishWitness F * a) ∧
    Continuous (fun a : JetA F => scottishWitness F * a) ∧
    IsStrictMap (fun a : JetA F => scottishWitness F * a) ∧
    IsClosed (Set.range
      fun a : JetA F => scottishWitness F * a) ∧
    (chartDatum F).canonicalMap (scottishWitness F) = 0 := by
  ...
```

For Problem 24, Lean uses the module structure induced by the canonical map to the completed localisation and proves `finiteJet_not_flat_canonicalMap`. The proof is the one given in Proposition 7.1: $Q^2$ is a non-zero-divisor of $A$, so flatness would make its image act injectively on the chart; that image is zero and the chart is nonzero. The ordinary ring localisation `Localization.Away` is always flat. The failure is a property of the completion defining `presheafValue`.

The conclusions of Lemma 5.1 are formalised for an arbitrary complete ultrametric normed commutative ring $E$ with a norm-scaling topologically nilpotent unit, subject to the noetherianity hypotheses stated there. The ring $T_E$ has its Gauss norm, $T_{E,0}$ is its unit ball, and the finite free terms have their sup norms. Positive-degree exactness is transported from the polynomial sequence by flat base change. Closedness in positive degree comes from exactness and continuity, whilst the degree-zero ideal is closed by the noetherian closed-ideal theorem. The closed-range theorem gives a constant $C_{q,E}$ such that every $y \in \mathrm{im}(d_{q,E})$ has a preimage $x$ with $\|x\| \leq C_{q,E}\|y\|$. This proves strictness, and multiplying such a preimage by a suitable power of $\varpi$ gives the two denominator inclusions (5) and (6).

The Lean proof of Lemma 5.2 uses these inequalities directly. Its localisation proof obtains the topological embedding from a quotient-norm estimate, and proves openness of $C_\alpha \to D_\alpha$ from the norm-preserving coefficientwise section. The components of Proposition 5.3 are therefore separate declarations. The transfer step is formalised for arbitrary complete nonarchimedean Huber rings carrying the local equaliser data of Theorem 6.1:

```
theorem isSheafy_of_milnorSquare
    (phiB : R →+* B) (phiC : R →+* C) (phiD : R →+* D)
    (hphiB : Continuous phiB) (hphiC : Continuous phiC)
    (hphiD : Continuous phiD)
    (sq : MilnorSquareData phiB phiC phiD hphiB hphiC hphiD)
    (hB : IsSheafy B) (hC : IsSheafy C) (hD : IsSheafy D) :
    IsSheafy R := by
  ...
```

The concrete finite-jet instance pushes finite rational covers to $B, C, D$ and supplies the required local rows and their compatibility with restriction.

The Koszul argument, the rational-chart calculation, and the transfer of sheafiness through the concrete Milnor square are contained respectively in the restricted-exactness file and the strictness and closed-image file, `FJP/Over/Chart.lean`, and `FJP/Over/SheafTransfer.lean`.

A.7. **The weighted-parity example.** The second development uses the same restricted-series infrastructure. Its ambient ring is the countable restricted Tate algebra

$$K\langle W, U_1, U_2, \ldots\rangle.$$

The definitions `wpWeight`, `WPMem`, and `wpSupport` are the direct translation of the formula for $\omega$ in (9), the definition of $S$ in (10), and the description of $\mathcal{A}$ in (13). The relevant part of the definition is as follows.

```
def wpWeight (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : ℕ :=
  ∑ n ∈ t.support,
    if (t n).mod 2 = 1 ∧ n ≠ 0 then w n else 0

def WPMem (w : ℕ → ℕ) (t : ℕ →₀ ℕ) : Prop :=
  wpWeight w t ≤ t 0

abbrev WPA : Type _ := ↑(wpSupport K w)
```

```
def idWeight : ℕ → ℕ := fun n => n

abbrev WPAid : Type _ := WPA K idWeight
```

Lean indexes the variables by $\mathbb{N}$ rather than $\mathbb{N}_{>0}$; the two conventions are equivalent here because the value `w 0` is ignored. The coordinate $t(0)$ is the exponent of $W$; the other coordinates are the exponents of the $U_n$. Thus `WPMem w t` says exactly that $a \geq \omega_w(\nu)$, and `wpSupport` requires the coefficients outside this monoid to vanish. It follows that `WPA K w` is the ring $\mathcal{A}_w$ of (13); `WPAid K` is the specialization $w(n) = n$. The maps `constA` and `norm_constA` give the isometric embedding of $K$ as the constant series. This map is explicit in the development, although it is not bundled as an `Algebra K` instance.

Suppose that $K$ is complete, nontrivially normed and ultrametric, let $w : \mathbb{N} \to \mathbb{N}$, choose a `Uniformizer K`, and assume the norm unit ball of $K$ is noetherian where indicated. The principal generic statements are:

```
theorem norm_wpa_mul (a b : WPA K w) :
    ‖a * b‖ = ‖a‖ * ‖b‖ := by
  ...

instance : IsDomain (WPA K w) := by
  ...

theorem isUniform_WPA (ϖ : Uniformizer K) :
    IsUniform (WPA K w) := by
  ...

theorem powerBoundedSubring_eq_unitBall (ϖ : Uniformizer K) :
    powerBoundedSubring (WPA K w) =
      (FiniteJet.unitBall (WPA K w) : Set (WPA K w)) := by
  ...

theorem not_isNoetherianRing_WPA
    (hw : ∀ n, 1 ≤ n → 1 ≤ w n) :
    ¬ IsNoetherianRing (WPA K w) := by
  ...

theorem wp_isSheafyComplete (ϖ : Uniformizer K)
    (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
    IsSheafyComplete (WPA K w) := by
  ...

theorem isDomain_chart (ϖ : Uniformizer K)
    (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
    IsDomain (presheafValue
      (chartDatum (w := w) ϖ)) := by
  ...

theorem not_isUniform_chart
    (hwu : ∀ M, ∃ n, 1 ≤ n ∧ M ≤ w n)
    (ϖ : Uniformizer K)
    (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
    ¬ IsUniform (presheafValue
      (chartDatum (w := w) ϖ)) := by
  ...
```

```
theorem not_isStablyUniform_WPA
    (hwu : ∀ M, ∃ n, 1 ≤ n ∧ M ≤ w n)
    (ϖ : Uniformizer K)
    (hK₀ : IsNoetherianRing (FiniteJet.unitBall K)) :
    ¬ IsStablyUniform (WPA K w) := by
  ...
```

The strong-sheafiness statement is in `WP/StrongSheafy.lean`:

```
theorem wp_tateExt_isSheafyComplete
    (ϖ : Uniformizer K)
    (hK₀ : IsNoetherianRing (FiniteJet.unitBall K))
    (s : ℕ) :
    letI := mvTateAlgebraTopology' (A := WPA K w) s
    haveI := mvTate_isTateRing (A := WPA K w) s
    haveI := mvTate_t2Space (A := WPA K w) s
    haveI := mvTate_nonarchimedean (A := WPA K w) s
    haveI := mvTateAlgebraTopology'_isTopologicalRing
      (A := WPA K w) s
    haveI : @CompleteSpace
      ↑(restrictedMvPowerSeriesSubring s (WPA K w))
      (IsTopologicalAddGroup.rightUniformSpace _) :=
        wp_tateExt_completeSpace (w := w) s
    IsSheafyComplete
      ↑(restrictedMvPowerSeriesSubring s (WPA K w)) := by
  ...
```

For every $w$, the restricted power-series subring is $\mathcal{A}_w\langle V_1, \ldots, V_s\rangle$. Thus this theorem is precisely the general strong-sheafiness assertion in Theorem 8.10; the separate `_of_dvr` endpoint packages the uniformizer and noetherianity hypotheses for a discretely valued base.

Here `IsSheafyComplete` is the predicate defined above: it quantifies over every ring of integral elements. In particular it proves the sheaf condition for $(\mathcal{A}, \mathcal{A}^\circ)$. The all-open theorem also records the corresponding statement directly for the structure presheaf. The datum `chartDatum` is $(W; \varpi)$, and `chartDatum_isRational` checks that it defines the chart $|W| \leq |\varpi|$. The chart is a domain for every $w$, and it is not uniform whenever $w$ is unbounded.

The proof routes are the ones used in section 8. Multiplicativity of the countable Gauss norm gives the domain and unit-ball statements. For rational data in $\mathcal{A}^{(N)}$, localisation is proved by taking the completed quotient of $\mathcal{A}^{(N)}$, applying its evaluation map to each coefficient in (16), and constructing the inverse by summing coefficients that tend to zero; it does not invoke the Koszul lemma. For sheafiness, one $N$ is chosen for the base datum and all members of a finite cover. Strong noetherianity of $\mathcal{A}^{(N)}$ glues each coefficient, and the restriction embedding shows that the resulting coefficients tend to zero. The final topological embedding is obtained by applying the nonarchimedean open mapping theorem over $K$ to the closed subspace of compatible tuples. Strong sheafiness is transported across the bicontinuous equivalence between a finite Tate extension and the corresponding weighted-parity algebra with shifted weight. For the bad chart, the quotient formed from $\mathcal{A}^{(0)} = K\langle W\rangle$ is $K\langle X\rangle$, and the coefficientwise ring embeds in formal multivariable power series over this domain. The formalisation also contains, under `WeightedParity.PerturbSetup`, a quantitative norm-unit-ball version of the standard perturbation result cited in Lemma 8.5.

These declarations, including the final failure of stable uniformity, have been checked to use only the usual logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

A.8. **Mathematical and code provenance.** The formal structure presheaf, rational-cover criterion, and the strongly-noetherian sheaf theorem follow Huber's construction [11, Theorem 2.2 and Lemmas 2.3–2.6] and Wedhorn's systematic formulation [21, Proposition and Definition 6.36, Remark 8.20, Definition 8.26, Theorem 8.28(b), Corollary 8.32, and Lemmas 8.33–8.34]. The passage between a fixed plus ring and all plus rings follows the standard rational-refinement argument recorded by Kedlaya [12, Lemma 1.6.8 and Remark 1.6.9]. Bambozzi–Kremnizer formulate the general construction in terms of Koszul complexes and prove strict Koszul regularity [3, Notation 4.3, Definitions 4.5–4.6, and Corollary 4.13]. The formalisation does not invoke this result. Positive-degree exactness is instead proved by flat base change from the polynomial sequence; closedness and the nonarchimedean open mapping theorem [14, Lemma 3.1] give constants bounding the norm of a preimage by a fixed multiple of the norm of its image. Multiplication by a suitable power of $\varpi$ then yields the bounded-denominator criterion used by Buzzard–Verberkmoes [4, Lemma 2].

The development is written in Lean 4 [15] on top of mathlib [16]. The formalisation-specific files live in AINTLIB [1]. AINTLIB includes code adapted from William Coram's restricted-power-series development [6]; its univariate Gauss-norm layer builds on Fabrizio Barroero's mathlib code [5]. It separately uses Bingyu Xia's multivariable power-series equivalences [9]. The vendored source headers and commit history retain the respective attributions. The finite-jet and weighted-parity definitions, localisation proofs, and sheaf endpoints discussed in this appendix are the AINTLIB-specific part of the development.

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